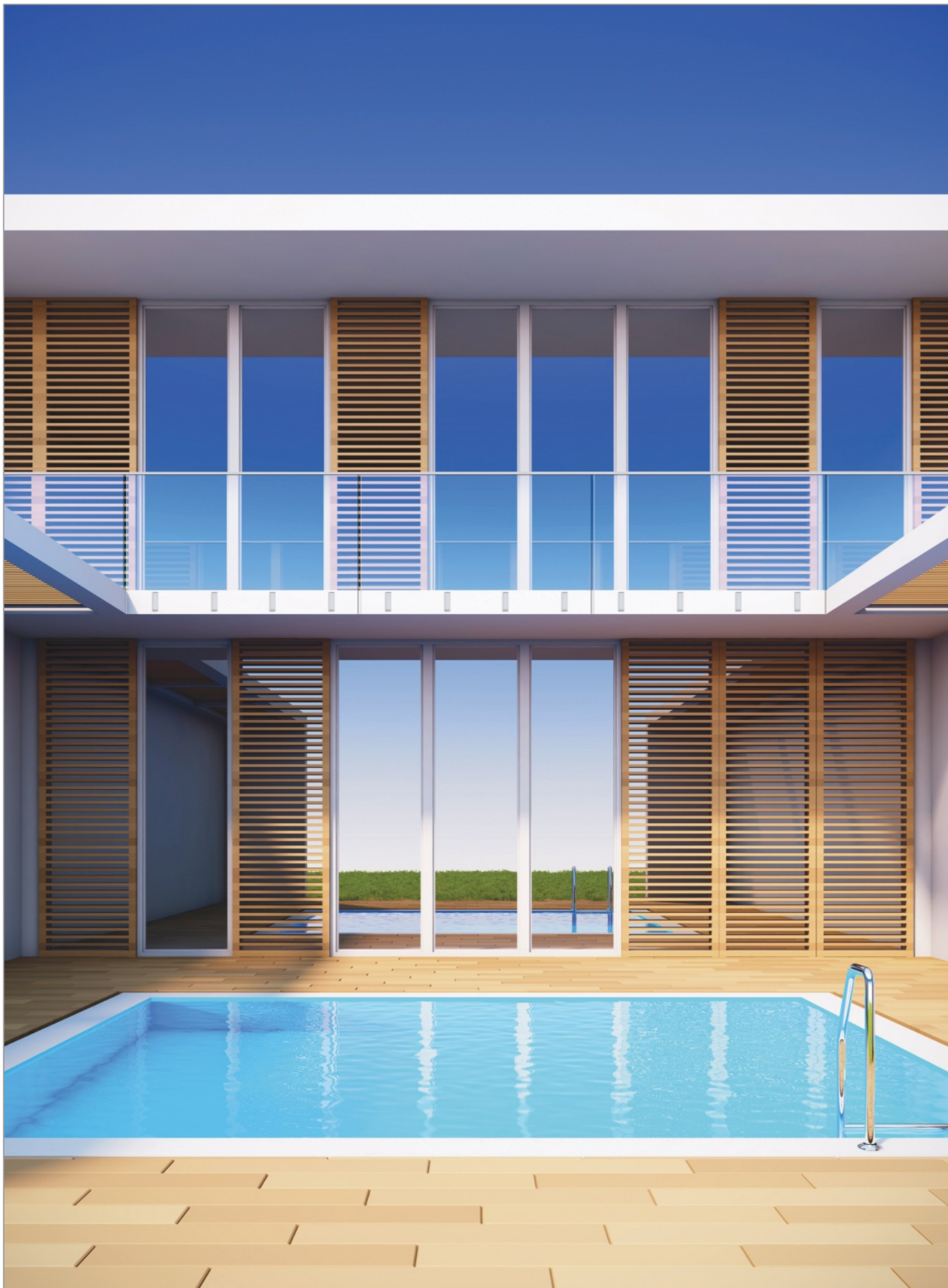




Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Edgetec JH Clamp® Balustrade
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	0.5-1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)

Note: If barrier extends more than 1.2m above fixing point, test reports must be provided

Supplier:	Royal Glass
-----------	-------------

PART 4: BARRIER DETAILS

System name:	Edgetec JH Clamp Balustrade
Drawings (ref page number)	
Engineers name:	Lautrec Technology Group
Hardware (type):	Aluminium Clamps

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)

- | | |
|--------------------------|---|
| ☐ | Continuous interlinking rail (show how fixed to building and each pane of glass) |
| ☐ | Continuous interlinking top cap (show how fixed to building and each pane of glass) |
| ☐ | Clamps, brackets or point fixings (show how fixed to building and each pane of glass) |
| ☐ | Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass) |
| ☐ | Glass with stiff interlayer |
| ☐ | Two or three glass edges supported by continuous clamps |
| ☐ | Structural post to fix interlinking rail to deck |
| ☐ | Other (specify) |

Engineers name:

Lautrec Technology Group

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz

NAME:

ADDRESS:

DATE:

CONSENT AUTHORITY: Auckland Council



THIS PS1 DOES NOT CONSTITUTE A SITE SPECIFIC REVIEW BY THE ENGINEERING COMPANY BELOW AND IS ONLY APPLICABLE FOR STANDARD FIXINGS AND DESIGN PARAMETERS AS DETAILED IN THE RELEVANT MANUAL



LAUTREC
CONSULTING ENGINEERS



association of
consulting and
engineering



PRODUCER STATEMENT – PS1 DESIGN

BUILDING CODE CLAUSE(S): B1, F2, F4, F9

JOB NUMBER: 1392-1803

ISSUED BY: Lautrec Technology Group Limited
(Engineering Design Firm)

TO: Juralco Aluminium Building Products Ltd
(Owner/Developer)

TO BE SUPPLIED TO: All Building Consent Authorities in NZ
(Building Consent Authority)

IN RESPECT OF: Juralco Edgetec® JH Clamp Balustrade System
(Description of Building Work)

AT: N/A (all locations within NZ Wind Zone)
(Address, Town/City)

LEGAL DESCRIPTION:

N/A ☒

We have been engaged by the owner/developer referred to above to provide (Extent of Engagement):

Specific engineering design for structural components only

in respect of the requirements of the Clause(s) of the Building Code specified above for Part only, as specified in the Schedule, of the proposed building work.

The design carried out by us has been prepared in accordance with:

- ☒ Compliance documents issued by the Ministry of Business, Innovation & Employment (Verification method/acceptable solution) B1/VM1 and/or;
- ☒ Alternative solution as per the attached Schedule.

The proposed building work covered by this producer statement is described on the drawings specified in the Schedule, together with the specification, and other documents set out in the Schedule.

On behalf of the Engineering Design Firm, and subject to:

- Site verification of the following design assumptions: assumed adequate support structure by others
- All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that:

- the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the Schedule, will comply with the relevant provisions of the Building Code and that;
- the persons who have undertaken the design have the necessary competency to do so.

I recommend the CM 1 level of construction monitoring.

I, (Name of Engineering Design Professional) Kevin Brown, am:

- ☒ CPEng number 140404

and hold the following qualifications BE, CMEngNZ, CPEng, IntPE(NZ), MBA

The Engineering Design Firm holds a current policy of Professional Indemnity Insurance no less than \$200,000

The Engineering Design Firm is a member of ACE New Zealand.

SIGNED BY (Name of Engineering Design Professional): Kevin Brown
(Signature below):

ON BEHALF OF (Engineering Design Firm): Lautrec Technology Group Limited

Date: 28/01/2026

Note: This statement has been prepared solely for the Building Consent Authority named above and shall not be relied upon by any other person or entity. Any liability in relation to this statement accrues to the Engineering Design Firm only. As a condition of reliance on this statement, the Building Consent Authority accepts that the total maximum amount of liability of any kind arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in tort or otherwise, is limited to the sum of \$200,000.

This form is to accompany **Form 2 of the Building (Forms) Regulations 2004** for the application of a Building Consent.

SCHEDULE to PS1

Please include an itemised list of all referenced documents, drawings, or other supporting materials in relation to this producer statement below:

Structural Maintenance Schedule refer to the relevant Juralco product manuals

B2 Letter

Engineering Drawings Refer to product manual Juralco Edgetec® JH Clamp Balustrade System

Engineering Calculations Refer to PS1 Report - Juralco Edgetec® JH Clamp Balustrade System_28.01.26

Alternative Solution

The design carried out by Lautrec Technology Group Limited has been prepared in accordance with the following alternative solutions:

NZS1170.2:2021 Incorporating Amendment No. 1

AS/NZS 1664.1:1997

NZS AS 1720.1:2022

NZS 3101:2006

NZS 3404:1997

EOTA TR 049 and EN1992-4:2018 for post-installed anchor design

NZS 4223.3:2016

NZS 4223.4:2008

NZS 8500:2016

28/01/2026

To the Building Official,

B2 COMPLIANCE - JURALCO EDGETEC® JH CLAMP BALUSTRADE SYSTEM

at *occupancy categories A, B, E, and C3*; at *any location in New Zealand that falls within the scope of this PS1* - see *supporting PS1 report for scope*

We have been asked to provide a PS1 for Clause B2 of the Building Code - Structural Durability

We are not able to provide this because there is no effective verification method for B2 contained within the New Zealand Building Code.

As these systems can be installed in a variety of settings, including internal and exposed environments, it is not deemed practical to specify durability requirements for the substructure. Timber treatments, mild steel corrosion protection coatings, and concrete and masonry covers are therefore up to the building designer to specify in accordance with the relevant recognised standards.

However, we can confirm that for the structural elements shown in the attached documentation:

Material	Means of compliance	Details
Juralco Edgetec® JH Clamp Balustrade System - Aluminium and Stainless Steel	Alternative Solution	Stainless steel components conform to SS316. Aluminium extrusions, components, and hardware conform to 6060 T5. Non-structural cladding has a durability requirement of not less than 50 years for this project. Note that the industry-proven life of Aluminium, Stainless Steel, and Glass would satisfy a 50-year design life requirement. We note that this is on a time-to-first-maintenance basis.
Connections - Stainless Steel fixings	Alternative Solution	All bolt and screw fixings within Juralco Edgetec® JH Clamp Balustrade System shall be minimum SS316. Non-structural fixings that are difficult to replace have a durability requirement not less than 50 years for this project. Note that the industry-proven life of Stainless Steel would satisfy a 50-year design life requirement.

It is assumed that these structural elements are fixed to adequate structures by others.

Minor tea staining may occur in coastal environments. Refer to Juralco Edgetec® JH Clamp Balustrade System manual issue 1-26 for the supplier's maintenance requirements.

Yours faithfully,

Managing Director
Kevin Brown
BE, CMEngNZ, CPEng, IntPE(NZ), MBA





Juralco Edgetec JH Clamp Semi Frameless Balustrade - Exterior Deck



Juralco Edgetec JH Clamp Semi Frameless Balustrade - Exterior Balcony

Juralco Aluminium Building Products Ltd designs and distributes specialist aluminium joinery systems through a national network of franchised fabricators and agents.

For more than 25 years we have been at the forefront of specialist aluminium door and window products suitable for New Zealand joinery and building methods. Our comprehensive product range includes security and insect screens, balustrades and gates, shutters and awnings, shower screens, wardrobe doors and organisers and internal doors.

The Juralco Edgetec® JH Clamp Balustrade system is designed for Frameless Glass, from 12mm to 17.52mm, Faced fixed and for Residential or Commercial use.

The system is extremely versatile, with 3 x clamp styles to choose from and can be used in a range of configurations to suit most modern architectural requirements.

Complies With AS/NZS 1170:2002, NZS 4223.3.2016, NZ Building Code B1, B2, F2, F4 and F9

JH Clamp Balustrade is for Domestic and Residential Occupancy types A, A Other and C3 and for Commercial Occupancy Types B, E and C3

Occupancy Types as per AS/NZ 1170.1.2002. Not suitable for Commercial C1/C2, C5 and D applications

Code	Type of Occupancy for part of the building or structure	Specific Uses	Glass
A	Domestic and Residential activities	All areas within or serving exclusively one dwelling including stairs, landings etc, but excluding external balconies and edges of roofs.	Residential, 12mm Toughened Glass, 15.2 mm Toughened Laminated or 13.52mm SentryGlas®
B, E	Offices and work areas not included elsewhere including storage areas.	Light access stairs and gangways not more than 600mm wide Fixed platforms, walkways, stairways and ladders for access Areas not susceptible to overcrowding in office and institutional buildings; also industrial and storage building.	Commercial, 15mm Toughened Glass, 17.2 mm Toughened Laminated or or 17.52mm SentryGlas®
A Other, C3	Areas without obstacles for moving people and not susceptible to over crowding	Stairs, landings, external balconies, edges of roofs etc.	Residential or Commercial as detailed above

**Note 1 All for 12mm, 15mm Toughened Glass, 15.2mm, 17.2mm Toughened Laminated Glass and 13.52, 17.52mm SentryGlas
All edges polished, all Holes to be smooth and chip and crack free**

Note 2 Juralco Balustrade Systems building code compliance documentation requires all balustrade installations are to be completed in accordance with the requirements of our authorised installer certification.

**Note 3 All Frameless Toughened Glass Balustrades must have an Interlinking Rail to conform to NZS 4223.3.2016
Stiffener Brackets are an option for Laminated and SentryGlas. Not Required for Swimming Pools**

Index

masterspec partner
Section 4852JB

Heading	Pages	Description.
Specifications	4	Juralco standard specification sheet and Powder coating recommendations
Configurations	5 - 7	Shows typical Balustrade layouts
	8 - 9	Shows typical Stair Layouts and attach details
General	10	Shows JH Clamp cross sections and all details
Components	11	Shows all Components, incl the 3 x different JH Clamps
Dimension Tables	12	Shows Dimension Tables and Design Loads
Mountings	13 - 23	Shows recommended Mounting details - to Timber (p13-165), Steel (p17-20) and Concrete (p21-22)
Glass Panels. Top edge. Safety	24 - 33	Shows all Interlinking Rail options (5 x Total). Swivel connectors and End connections
	34	Shows Glass stiffener brackets, Fixed and Adjustable angles
End Brackets	35 - 37	Shows Interlinking Rail End bracket attachments to Posts and Structures
Rail mounted to Glass Panel	38	Shows Interlinking Rail Side mounted to a Glass Panel and all End brackets
Joiners and End Plates	39 - 40	Shows Interlinking and Handrail Joiners and End Plates
Surface Care	41 - 43	Instructions for the care of Glass, Powder Coated and Stainless Steel surfaces

Juralco Edgetec® JH Clamp Balustrade System - Specifications, Powder Coating

Juralco Aluminium Building Products Ltd (JABP) Specifications for Juralco Edgetec® JH Clamp Balustrade System

1. Scope

- This specification details the documents the Juralco Edgetec® JH Clamp Balustrade System refers to in relation to the New Zealand Building Code, the manufacturer's documents, products used in the System, requirements in relation to fixing and surface finishing.

2. NZBC Compliance

- The Juralco Edgetec® JH Clamp Balustrade System has been reviewed by Lautrec Technology Group Ltd to demonstrate compliance with the structural requirements of the New Zealand Building Code and NZS 1170 : 2002 occupancy A, B, E ,A Other and C3 to NZS 3604 up to and Including Extra High Wind Zone. 2.5kPa
- The Structural Engineering design includes the requirements of B1 Structure, B2 Durability, F2 Hazardous material and F4 Safety from falling, all from the Building Code.
- Verification Method B1 / VM1, B2/AS1, F4 / AS1
- All glass used in the Juralco Edgetec® JH Clamp Balustrade System must conform to AS/NZS 2208. Complies with NZS 4223.3.2016
- Separation of dissimilar materials (as relates to B2 compliance) have been reviewed.
For other combinations refer to NZS 3604:2011 Section 2.3.3 Separation and Section 4 Durability

3. Manufacturer's Documents

- The Juralco Edgetec® JH Clamp Balustrade System manual details all extrusions and components used for the fabrication and installation/fixing of the system.
- A Producer Statement 1(Design) is available.
Copies of the above documents are available from:
Juralco Aluminium Building Products Ltd
48 Bruce McLaren Rd, Henderson, Auckland
Phone 09 478 8018 Fax 09 478 7883 Email specify@juralco.co.nz
- Any deviation from the standard fabrication or installation/fixing must be accompanied by a site specific PS1 with site specific calculations and drawings. Contact Juralco for further information.

4. Products

- Only extrusions, components and hardware supplied by or specified by JABP may be used in the Juralco Edgetec® JH Clamp System
- Aluminium extrusions, components and hardware – unless specified are manufactured to 6060 T5 specifications
- Stainless Steel components, hardware, fixings – all components to 316 grade
- Glass - all glass used in the Juralco JH Clamp Balustrade System must conform to the specifications as listed in the Juralco JH Clamp manual with each panel conforming to AS/NZS 2208 as confirmed by the Safety Stamp detailing the manufacturer's description and licence number.

5. Surface Finishing

- Juralco Aluminium Building Products Ltd is a Dulux Registered Applicator site, registration number 2101.
JABP uses only Dulux branded powder coating materials
- Unless specified otherwise, Dulux Duralloy® powder coating systems are used for properties greater than 100 metres from high tide level where AAMA 2603 performance is required
- Dulux Duratec® powder coating systems must be used for all properties greater than 10 metres and up to 100 metres from high tide level where AAMA 2604 performance is required
- Dulux Duralloy® has a 10 year film and colour integrity warranty, Dulux Duratec® has a 20 year film and colour integrity warranty

6. Installation and Fixing

- The Juralco Edgetec® JH Clamp Balustrade System must only be installed in accordance with the Juralco Edgetec® JH Clamp Balustrade System manual
- Any deviation from that specified in the Juralco Edgetec® JH Clamp manual must only be in accordance with the site specific PS1 with site specific calculations and drawings listing the non standard details
- The Juralco Edgetec® JH Clamp Balustrade System must only be fabricated/installed by a Juralco approved fabricator
- Upon completion of the installation the fabricator must supply the owner with a PS3 (Construction)

Important information - Powder Coating systems.

Powdercoat Systems The new standard Dulux powder coating system used by Juralco is Duralloy Plus®. Also Duralloy® and Duratec®. All as per specs above. Juralco Powder coated prices are for Duralloy Plus® and Duralloy® (same pricing). Duratec® prices on application.

Attachment to structures A PVC Tape or similar material spacer must be used to separate powder coated aluminium items from all concrete and steel structures. Failure to do so can lead to the chemicals in the structure affecting the powder coating, leading to corrosion.

Swimming Pools The chlorinated water in swimming pools can cause the deterioration of powder coated surfaces, leading to corrosion of the underlying surface. It is recommended that Powder coated surfaces be 1200mm min from a pool.

Care The Dulux powder coating warranty period is conditional upon the surface being maintained in accordance with the Dulux 'Care and Maintenance Instructions'. Download from Dulux or refer to the back page of this manual.



Juralco Edgetec® JH Clamp Balustrade System - Typical Layouts

JH Clamp + Interlinking Rail

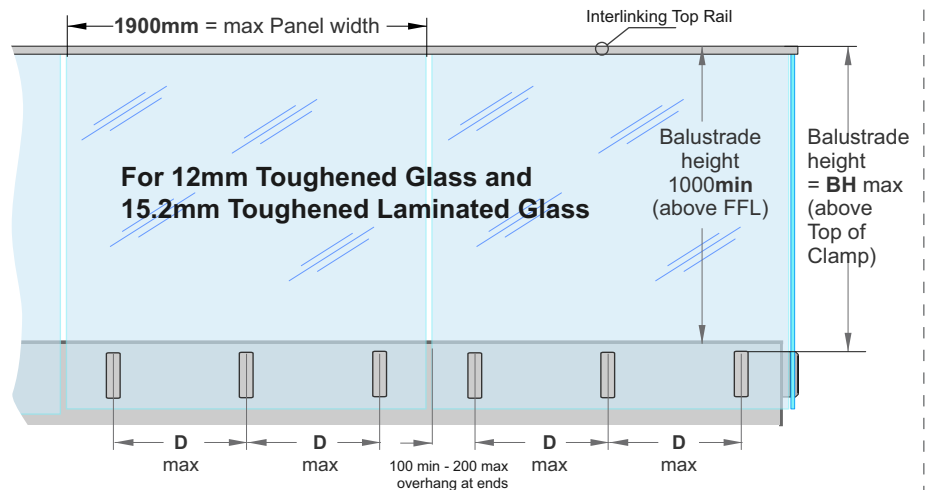
Glass must have a minimum strength of 100MPa. All edges polished

Residential & Domestic only
Occupancy types A, A Other and C3

Extra High Wind Zone

12mm Toughened Glass and
15.2mm Toughened Laminated Glass

Note: See individual Mounting
pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

JH Clamp + Interlinking Rail

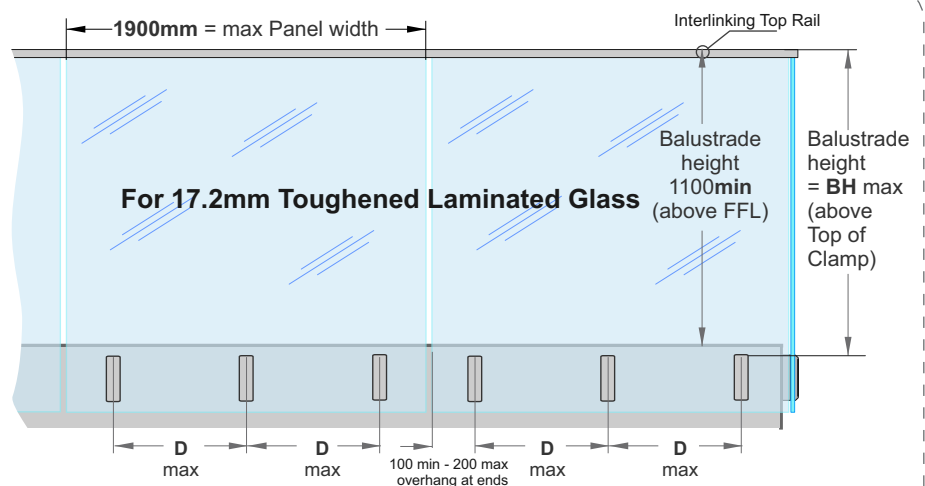
Glass must have a minimum strength of 100MPa. All edges polished

Commercial Occupancy
types B, E, and C3 only

Very High Wind Zone

17.2mm Toughened Laminated Glass

Note: See individual Mounting
pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Very High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

JH Clamp + Interlinking Rail

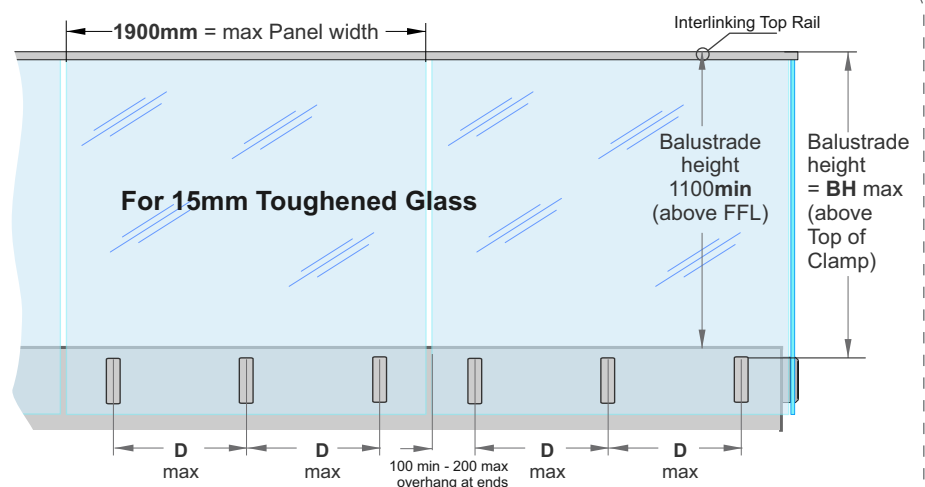
Glass must have a minimum strength of 100MPa. All edges polished

Commercial Occupancy
types B, E, and C3 only

Extra High Wind Zone

15mm Toughened Glass

Note: See individual Mounting
pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

Juralco Edgetec® JH Clamp Balustrade System - Typical Layouts

JH Clamp + Stiffener Brkts

Glass must have a minimum strength of 100MPa. All edges polished

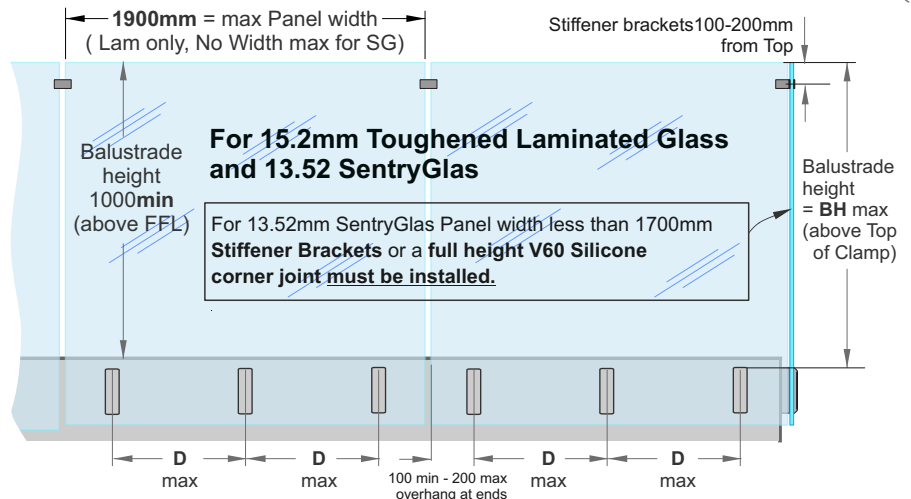
Residential & Domestic only
Occupancy types A, A Other and C3

Extra High Wind Zone

15.2mm Toughened Laminated Glass

13.52mm SentryGlas

Note: See individual Mounting pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

JH Clamp + Stiffener Brkts

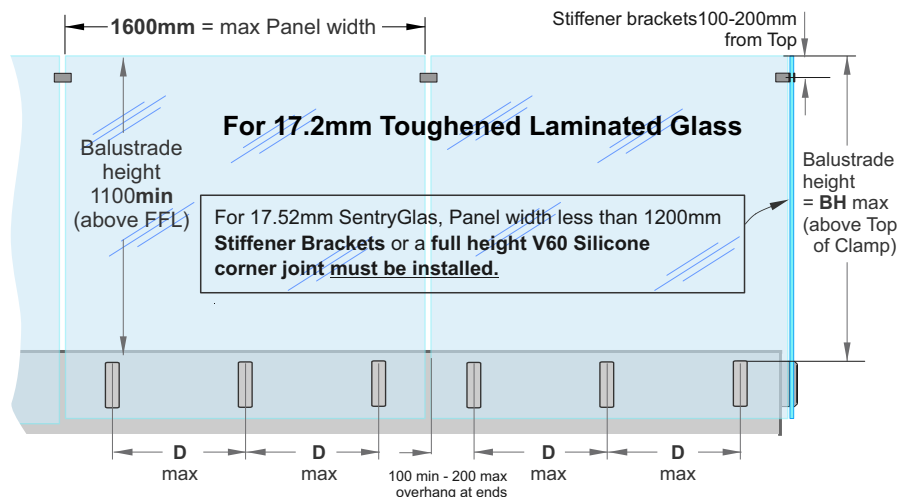
Glass must have a minimum strength of 100MPa. All edges polished

Commercial Occupancy
types B, E, and C3 only

Very High Wind Zone

17.2mm Toughened Laminated Glass

Note: See individual Mounting pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Very High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

JH Clamp + Stiffener Brkts

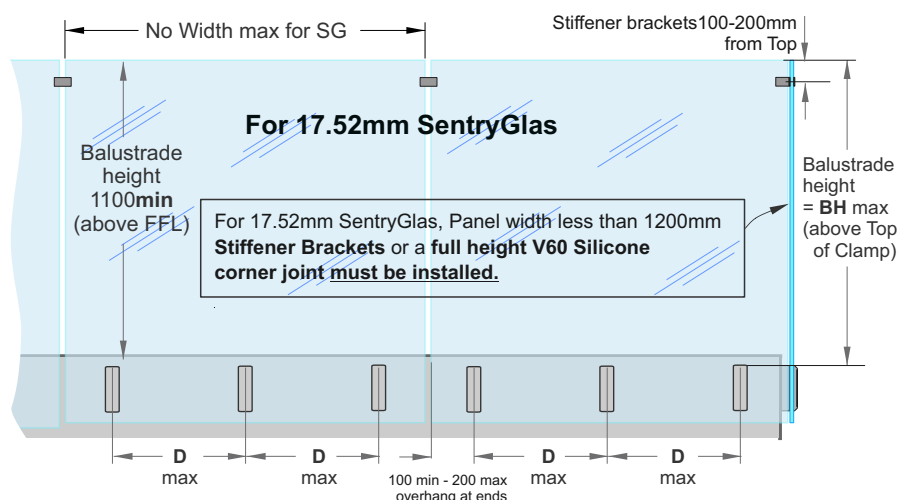
Glass must have a minimum strength of 100MPa. All edges polished

Commercial Occupancy
types B, E, and C3 only

Extra High Wind Zone

17.52mm SentryGlas

Note: See individual Mounting pages for construction options



Exceeds the wind loading for all Wind Zones up to **and including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

Juralco Edgetec® JH Clamp Balustrade System - Typical Layouts

JH Clamp Pool Fencing

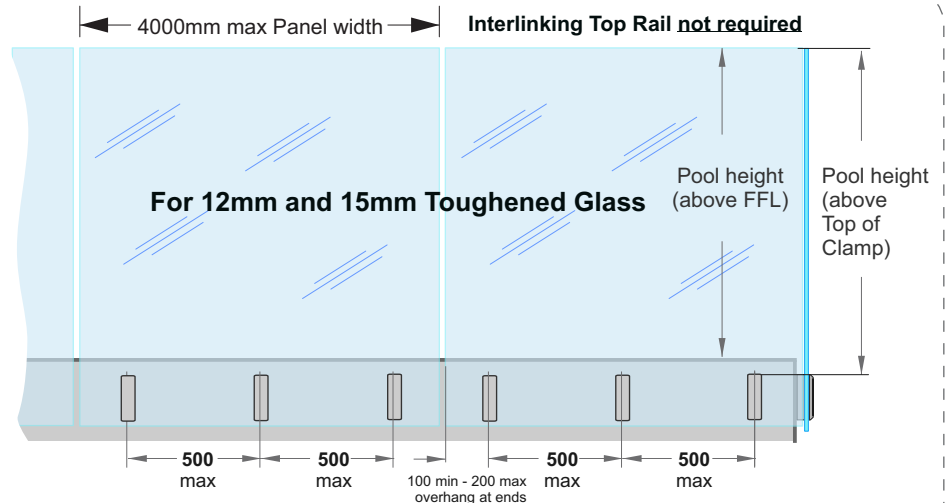
POOL FENCING only

Glass must have a minimum strength of 100MPa. All edges polished

Applies to Swimming Pools as of Jan 2017, complies with the Building Code clause F9 and section 162C of the Building Act.

Applies to Pool Fences not protecting a fall of 1.0m or more

Note: See individual Mounting pages for construction options



SentryGlas® Glass Layers and Thickness Orientation

Glass Thickness (mm)	Inner Layer of Glass thickness (mm) Deckside	Interlayer thickness(mm) and Type	Outer Layer Glass thickness (mm)
13.52	6	1.52 SentryGlas®	6
17.52	8	1.52 SentryGlas®	8

Refers to previous page. Laminated Glass Layers and Thickness Orientation

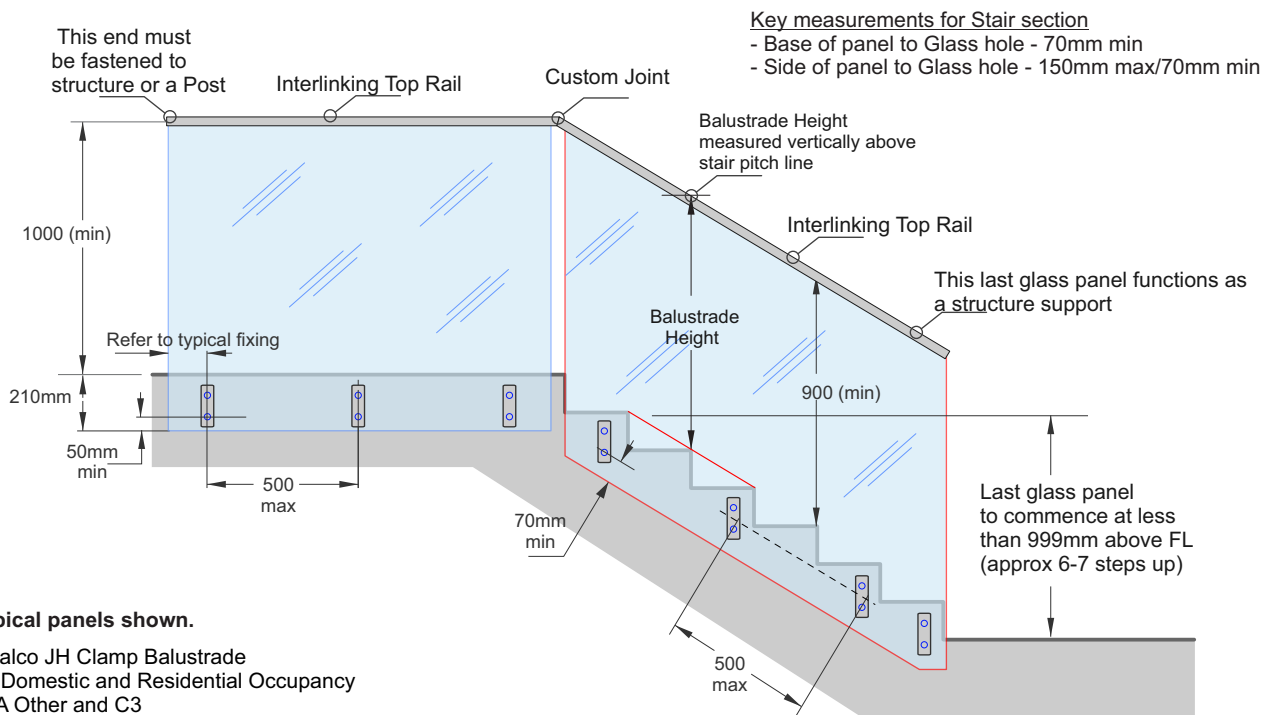
Glass Thickness (mm)	Inner Layer of Glass thickness (mm) Deckside	Interlayer thickness(mm) and Type	Outer Layer Glass thickness (mm)
15.2	8	1.2EVA	6
17.2	8	1.2EVA	8



Juralco Edgetec® JH Clamp Balustrade System - Typical Stair Setout

JH Glass Clamps Stairs - Vertical

Stair structure to be designed by others to resist Balustrade actions as per NZS1170.1 Table 3.3



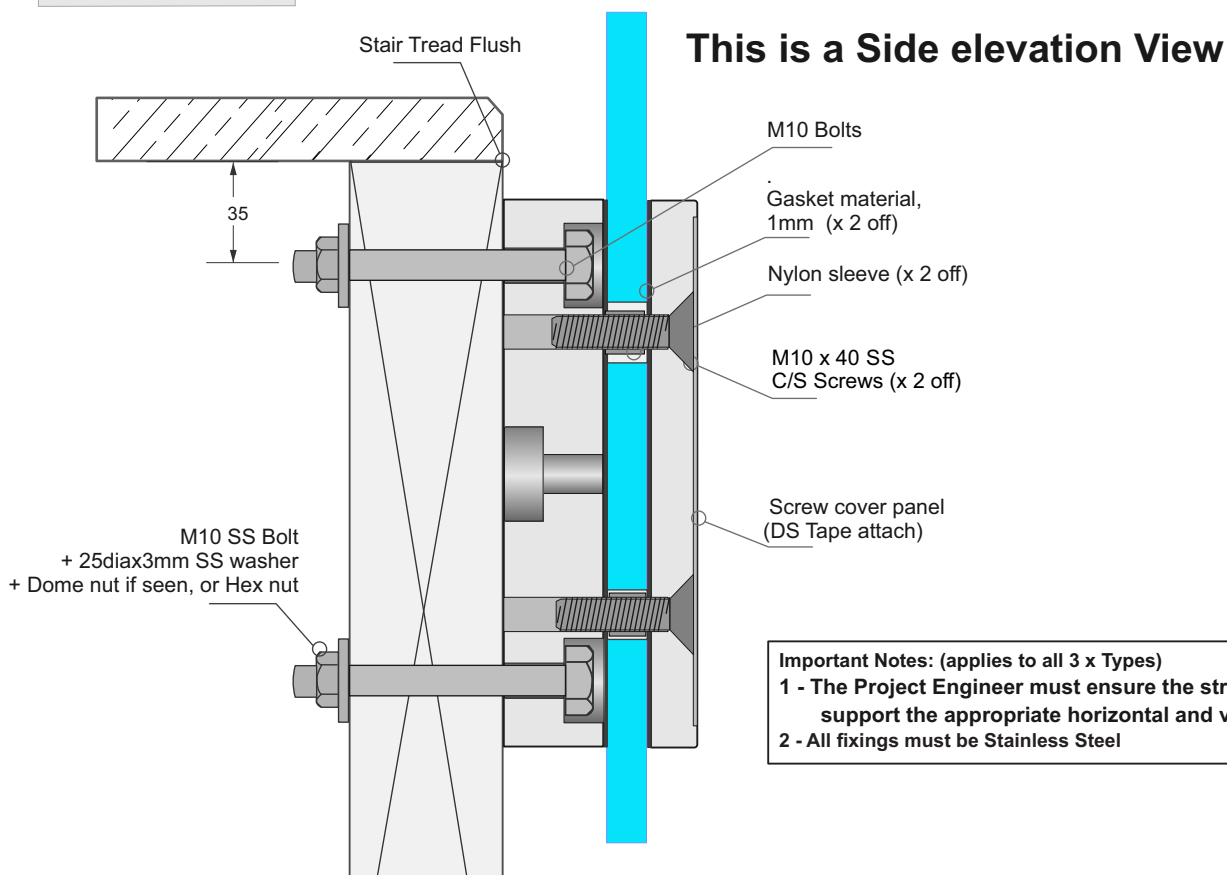
Typical panels shown.

Juralco JH Clamp Balustrade
for Domestic and Residential Occupancy
A, A Other and C3

Glass must have a minimum strength of 100MPa
All edges polished
All Holes to be smooth and chip and crackfree

JH Clamp Balustrade Stair Stringer Detail JET/JH165/SF shown

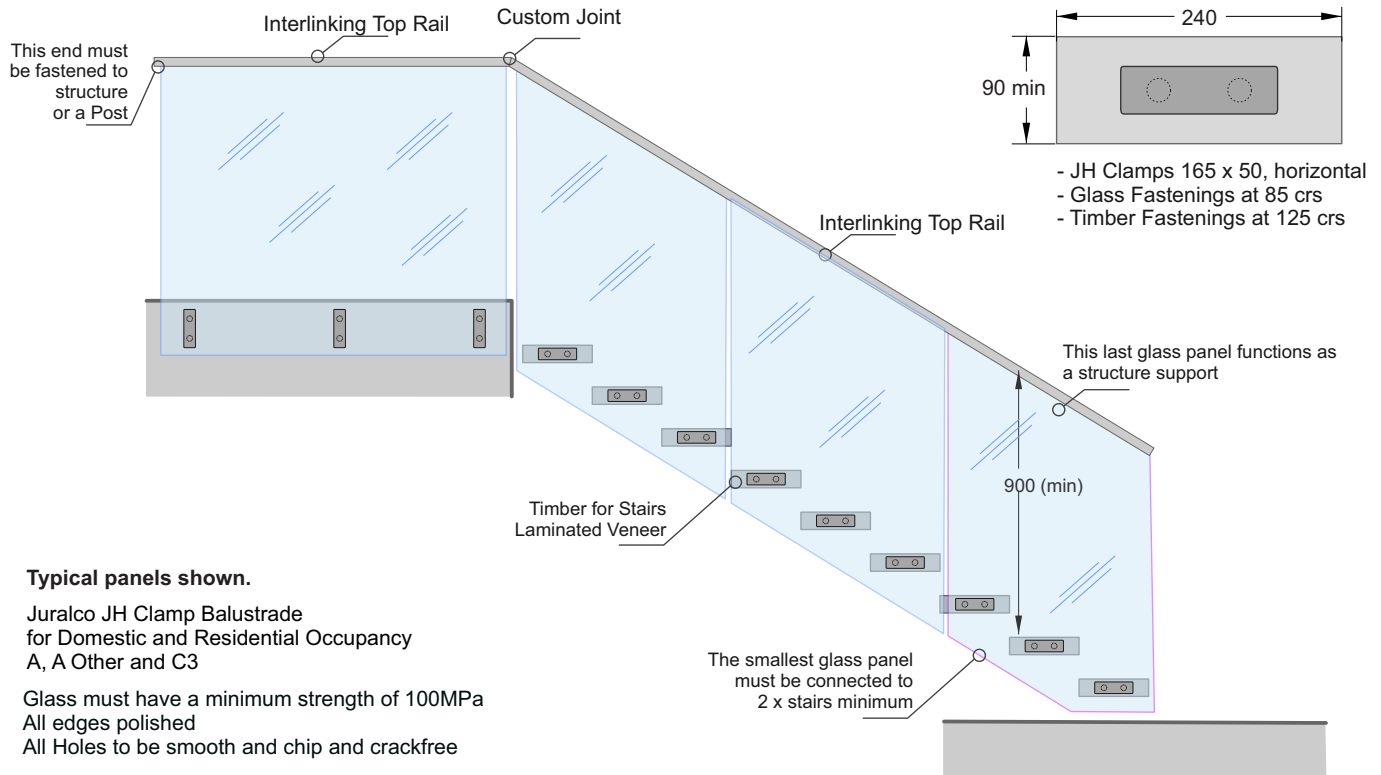
Stair structure to be designed by others to resist Balustrade actions as per NZS1170.1 Table 3.3
For Domestic and Residential Occupancy A,B,E



Juralco Edgetec® JH Clamp Balustrade System - Typical Stair Layout

JH Glass Clamps Stairs - Horizontal

Typical Floating Timber Stair, 240mm tread, 180mm step.
Landing area - Screw fixings into Single or Double Joists/Stiffeners, Concrete or Steel
Timber for Stairs - Laminated Veneer, 240x90mm min.



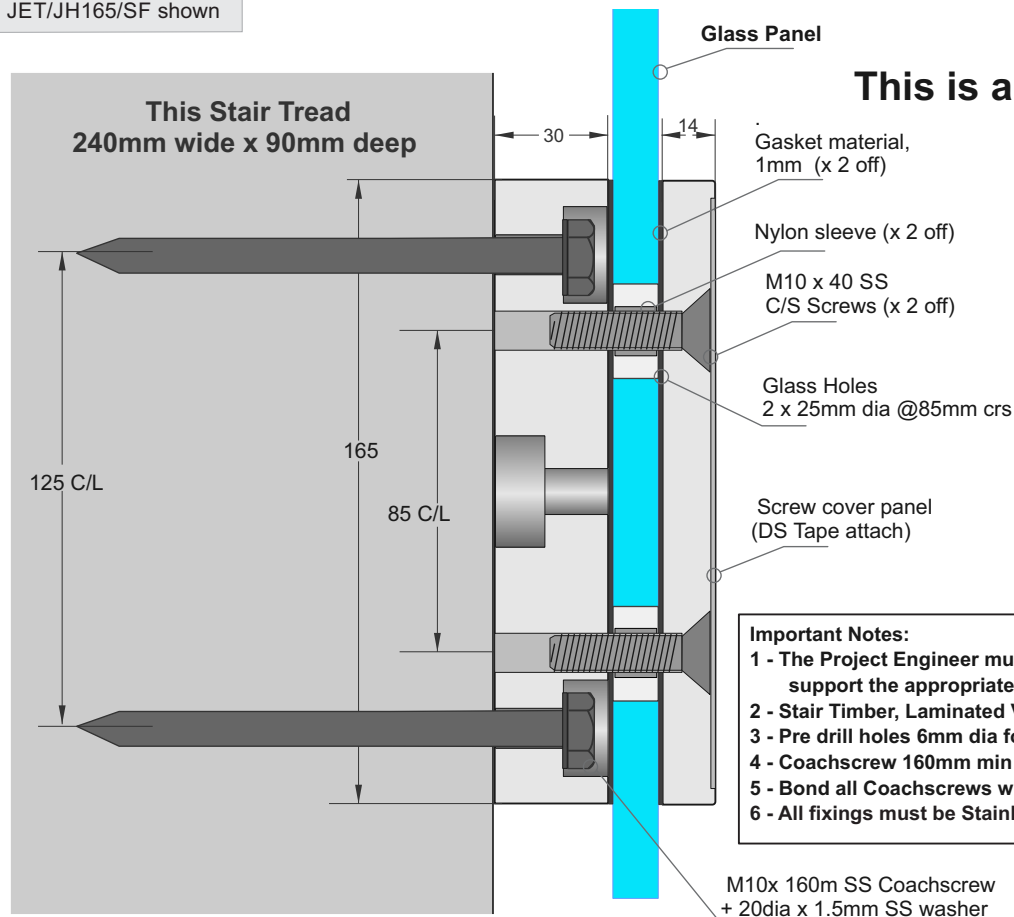
Typical panels shown.

Juralco JH Clamp Balustrade
for Domestic and Residential Occupancy
A, A Other and C3

Glass must have a minimum strength of 100MPa
All edges polished
All Holes to be smooth and chip and crackfree

JH Clamp Balustrade Stair Tread Detail JET/JH165/SF shown

Stair structure to be designed by others to resist Balustrade actions as per NZS1170.1 Table 3.3



Important Notes:

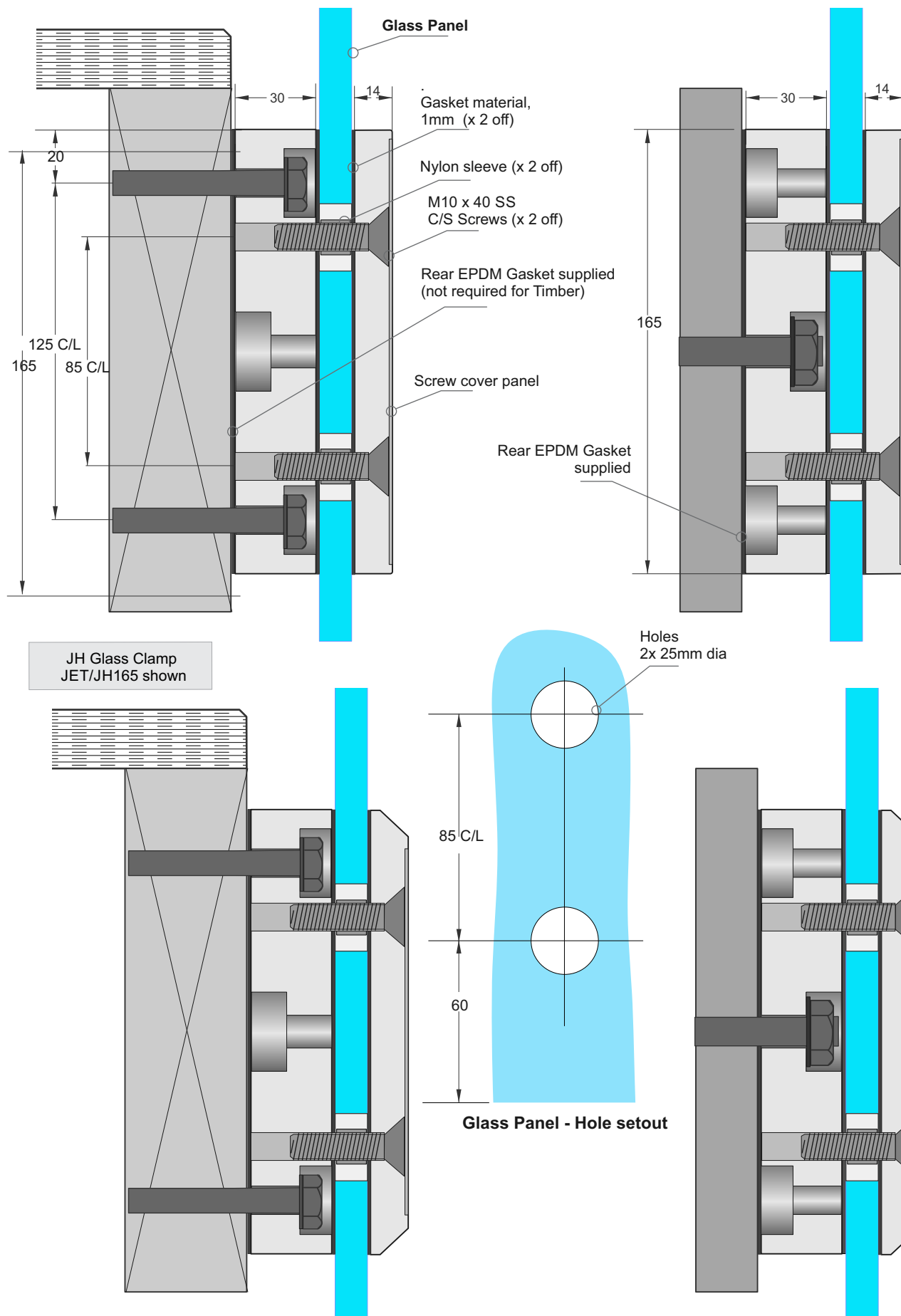
- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Stair Timber, Laminated Veneer
- 3 - Pre drill holes 6mm dia for Coachscrews .
- 4 - Coachscrew 160mm min screw engagement into Stair
- 5 - Bond all Coachscrews with SIKa Supergrip30 to full depth
- 6 - All fixings must be Stainless Steel

Juralco Edgetec® JH Clamp Balustrade System - Dimensions

JH Glass Clamp
JET/JH165/SF
or JET/JH165R
shown

Elevation showing the Main Features.
Rear Clamp Block in Timber position
**2 x M10 SS, Bolts or Coachscrews
for Fixing to Timber**

Elevation showing the Main Features
Rear Clamp Block in Steel position
**1 x M12 SS Stud
For Fixing to Steel or Concrete**

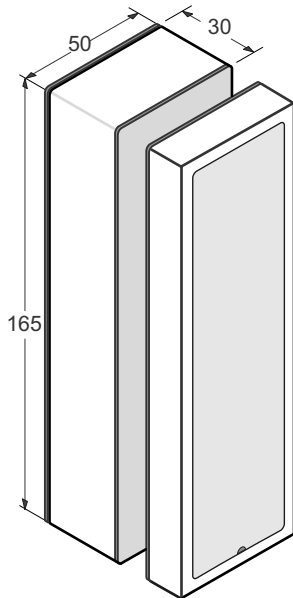


Juralco Edgetec® JH Clamp Balustrade System - Components

JH Clamp Mount JET/JH165SF

If accessible and
Top of Clamp
is between

150mm and 760mm above FFL, a Toehold is
present and this unit cannot be used.
Refer F4/AS1 for further information

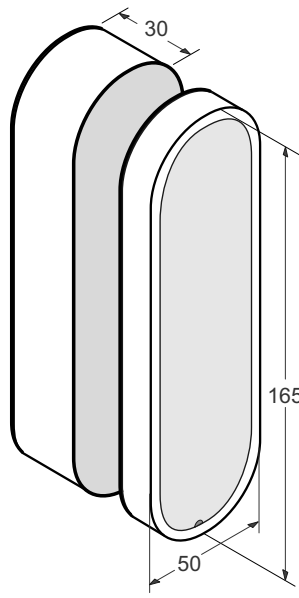


Comes as a kit, incl 2 x M10x55
SS C/S Machine screws,
3 x Gaskets and the Cover Plate
Cover Panels held in place
with JMF/X02 Double sided tape

JH Clamp Mount JET/JH165R

If accessible and
Top of Clamp
is between

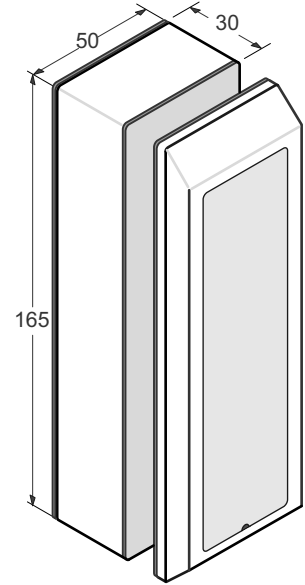
150mm and 760mm above FFL, a Toehold is
present and this unit cannot be used.
Refer F4/AS1 for further information



Comes as a kit, incl 2 x M10x55
SS C/S Machine screws,
3 x Gaskets and the Cover Plate
Cover Panels held in place
with JMF/X02 Double sided tape

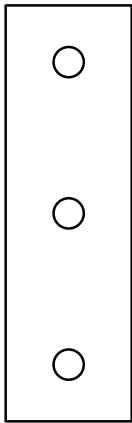
JH Clamp Mount JET/JH165

F4/AS1 Toehold does not apply



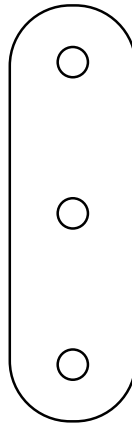
Comes as a kit, incl 2 x M10x55
SS C/S Machine screws,
3 x Gaskets and the Cover Plate
Cover Panels held in place
with JMF/X02 Double sided tape

JH Clamp Square Packer JET/JH165SB/CSP/1.0 - 1mm Packer JET/JH165SB/CSP/5.0 - 5mm Packer



165 x 50mm

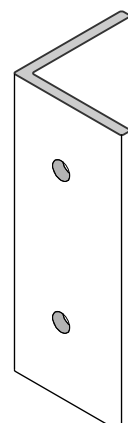
JH Clamp Round Packer JET/JH165RB/CSP/1.0- 1mm Packer JET/JH165RB/CSP/5.0 - 5mm Packer



165 x 50mm

Angle - Hidden Fixing JET/JH165HFBKT

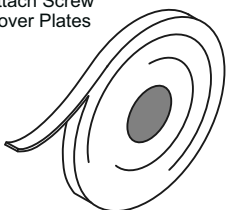
Kit includes 2 x M10 x 55 SS C/S screws



Angle ex 75x75x6 x165 long

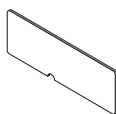
Acrylic Tape 8mm x 0.8mm JMF/X02/66

Attach Screw
Cover Plates

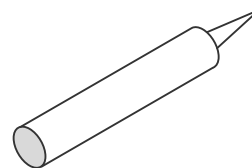


Double sided 8mm wide x 66mt Roll

Glass Bracket Replacement Cover Plate JET 68/51

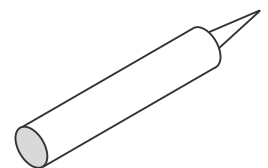


SIKA Supergrip JECSUPERGRIP30



For All Coachscrews fixings

Rhodorsil V60 Clear Silicone H/RTV419098



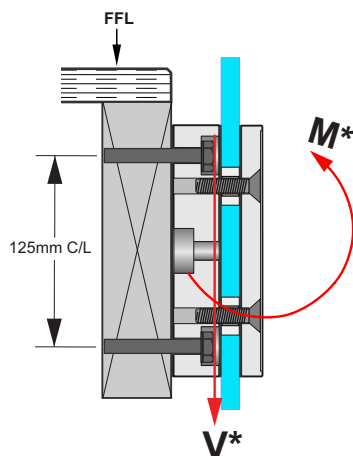
Construction Silicone

Juralco Edgetec® JH Clamp Balustrade System
Dimension Tables and Design Loads

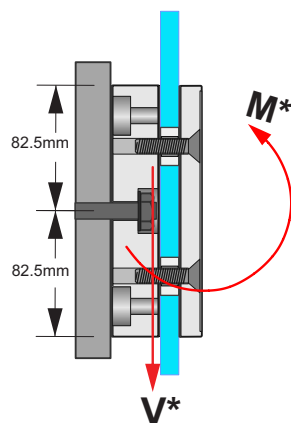
Occupancy A, A Other and C3 - Residential						
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)	M*(kNm)	V*(kN)
12T, 13.52SG, 15.2L	Extra High	1050	400	200	0.62	1.12
	Very High	1150	500		0.76	1.26
12T	Very High	1250	400		0.71	1.09

Occupancy B, E and C3 - Commercial						
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang	M*(kNm)	V*(kN)
17.2L	Very High	1250	400	200	0.86	1.32
17.52SG	Extra High	1250	500		1.07	1.65
15T	Extra High	1350	400		1	1.42

Pool Fence Only						
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang	M*(kNm)	V*(kN)
12T, 13.52SG	Medium	1400	500	200	0.6	0.84
	High	1350			0.8	1.14
	Very High	1275			0.93	1.39
15T, 17.52SG	Medium	1450			0.65	0.86
	High	1400			0.86	1.18
	Very High	1350			1.03	1.47
	Extra High	1275			1.12	1.68



**2 x M10 SS, Bolts or Coachscrews
for Fixing to Timber**

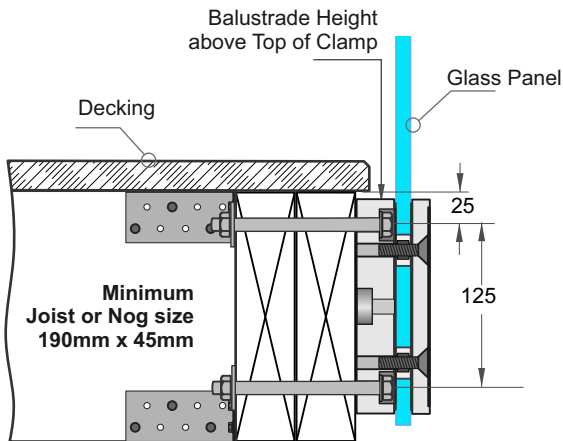


**1 x M12 SS Stud
For Fixing to Steel or Concrete**

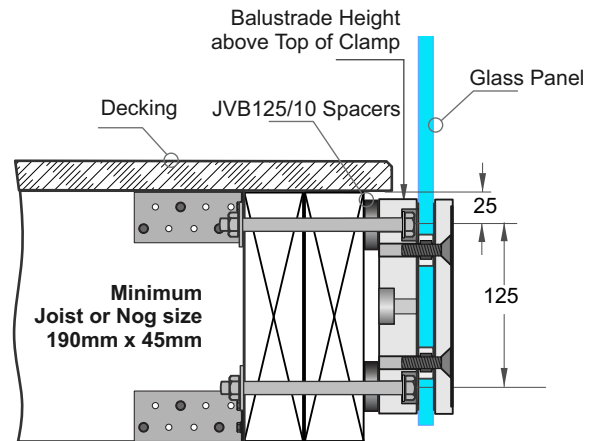
Glass Type			
Glass Thickness (mm)	Interlayer Thickness & Type	Minimum Panel Width (mm)	Maximum Panel Width (mm)
12T	-	200	1900
13.52SG	6/1.52/6	1700 with Stiffeners or 200 without Stiffeners	See manufacturers limits
15T	-	200	1900

- The Project Engineer must ensure the structure can support the appropriate loads
- Design loads can be used for Project Engineer to verify non standard fixing method and support structures.
- Max 60mm spacing back to structure including JVB125 Spacer
- For 13.52 SG Stiffener Brackets or Handrail must be used when above 1050mm BH or width is below 1700mm for barrier protecting a fall - not required for Pool Fence ONLY
- Maximum 10mm tolerance allowed to Barrier Height
- Minimum Glass strength 100MPa, all edges polished
- See "Interlinking Rail Manual" for maximum interlinking spans

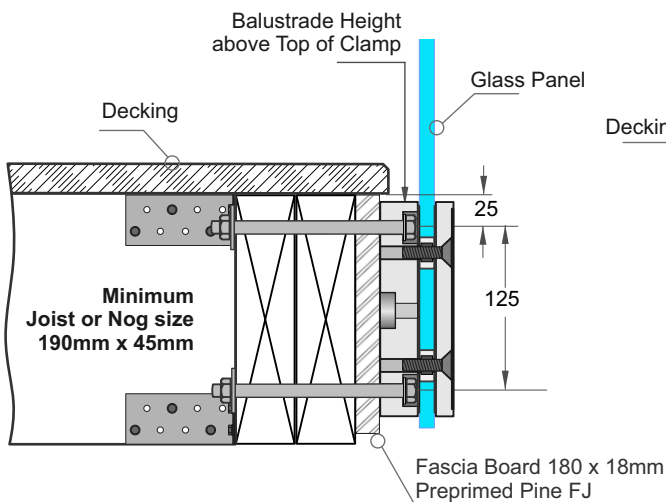
Approved Timber Construction Options
Face Fix into Double Joists (applies to all 3 x types)
M10 SS Bolts or Threaded Rod - All Wind Zones
M10 SS Coachscrews - All Wind Zones
Note: All Coachscrews full Screw engagement into Joists



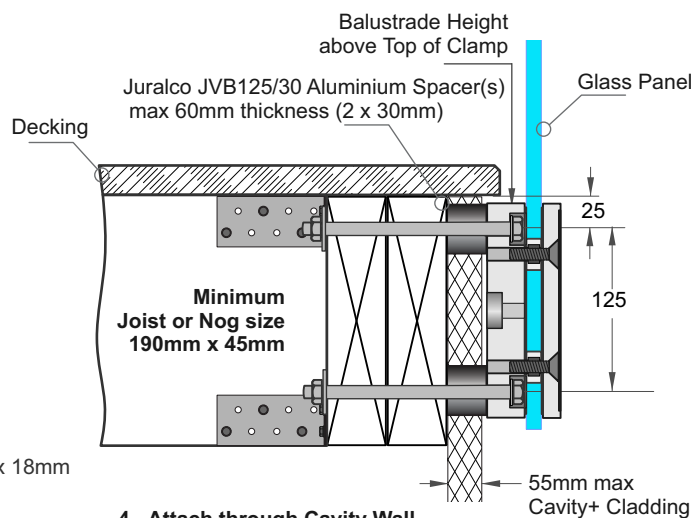
1 - Attach Directly to Double Joists



2- Attach Directly to Double Joists using 10mm Spacers.



3 - Attach Directly to a Fascia then to Double Joists



4 - Attach through Cavity Wall to Double Joists using 30mm Spacers (60mm max)

Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing
Complies with NZS3604:2011 - Double Boundary Joists

Typical FACE Fix to Timber - M10 SS Coachscrews

Occupancy A, A Other and C3 - Residential

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial

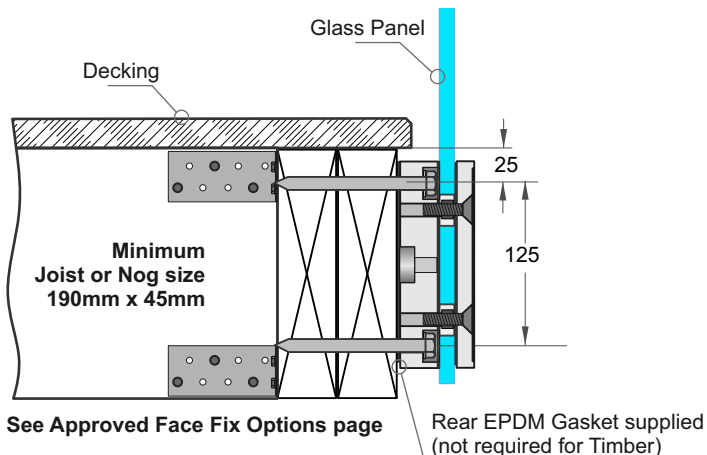
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

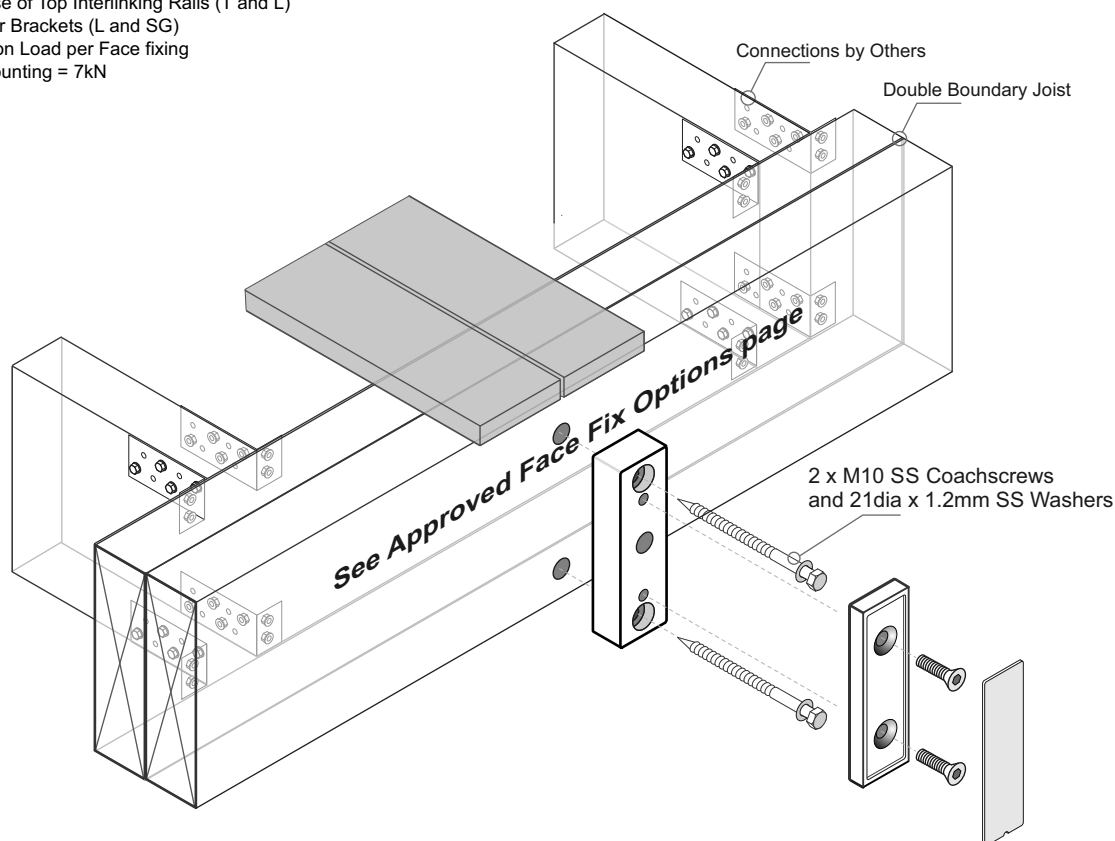
General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - Coachscrews 90mm min screw engagement into joists. Drill 6mm hole
- 4 - Bond all screws with SIKa Supergrip to full depth
- 5 - All fixings must be Stainless Steel
- 6 - Max 60mm spacing back to structure including JVB125 Spacer



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing
Complies with NZS3604:2011 - Double Boundary Joists

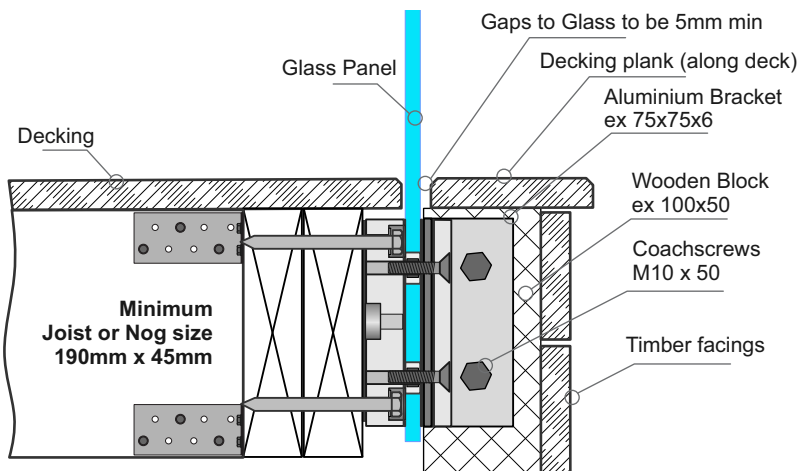
Typical HIDDEN FACE Fix to Timber - M10 SS Coachscrews

Note: Decking Planks and Timber Facings must be screw attached to allow for removal under B2 Durability requirements

Occupancy A,A Other and C3 - Residential				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

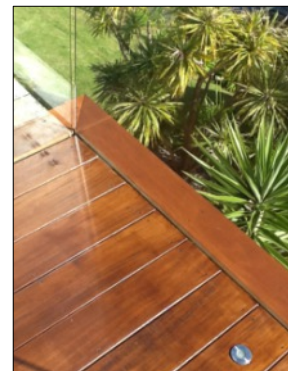
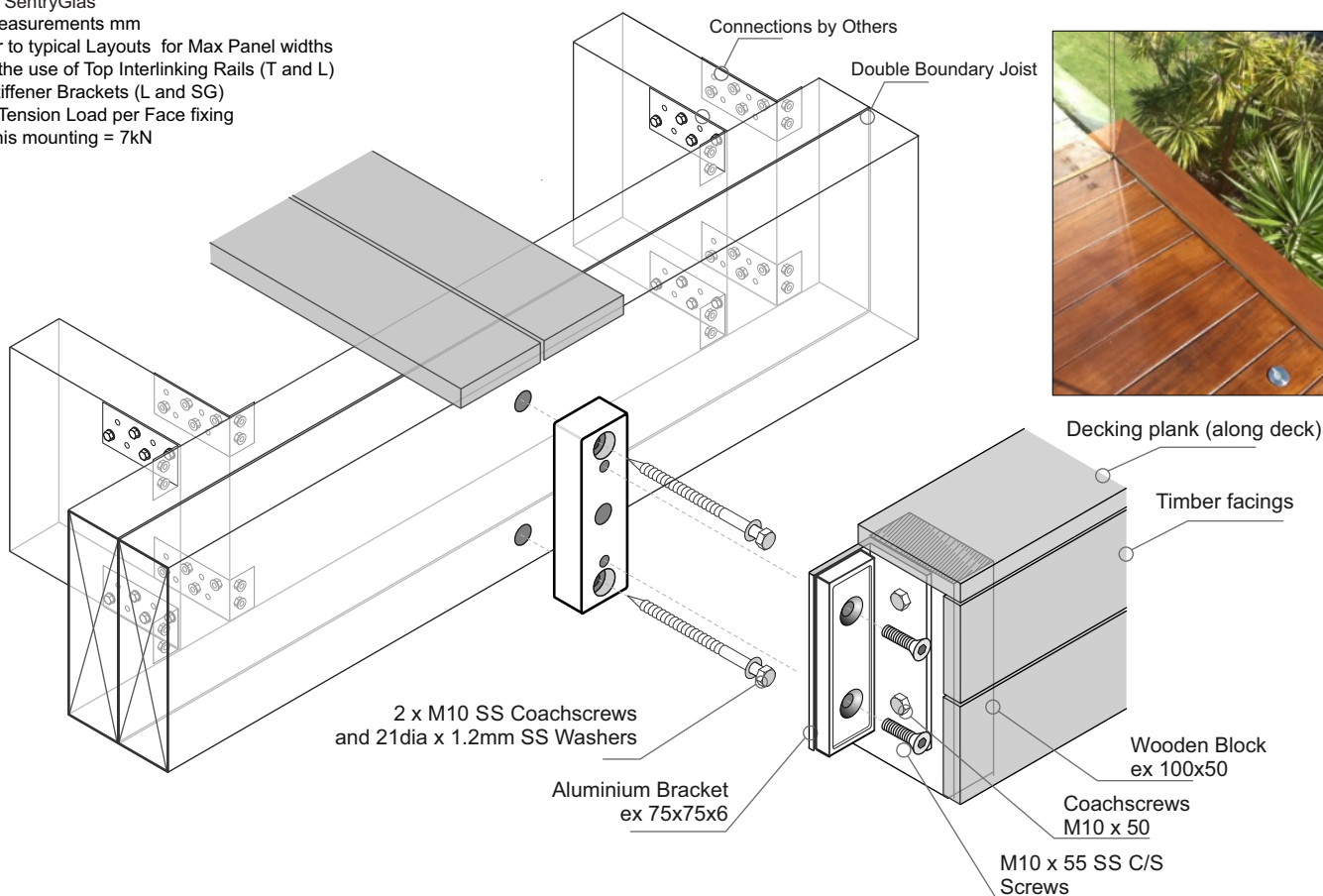


Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - Coachscrews 90mm min screw engagement into joists. Drill 6mm hole
- 4 - Bond all screws with SIKa Supergrip to full depth
- 5 - All fixings must be Stainless Steel
- 6 - Max 60mm spacing back to structure including JVB125 Spacer

General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing
Complies with NZS3604:2011 - Double Boundary Joists

Typical FACE Fix to Timber - M10 SS Bolts

Occupancy A, A Other and C3 - Residential

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial

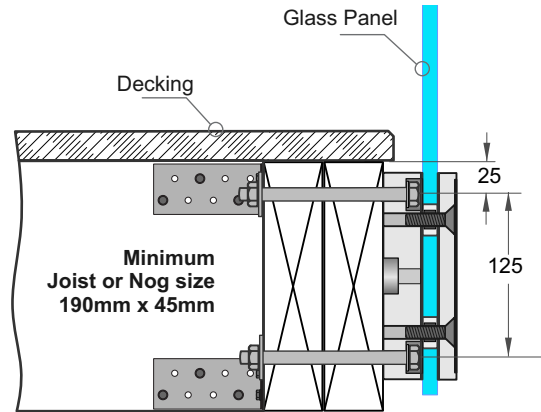
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

General Notes:

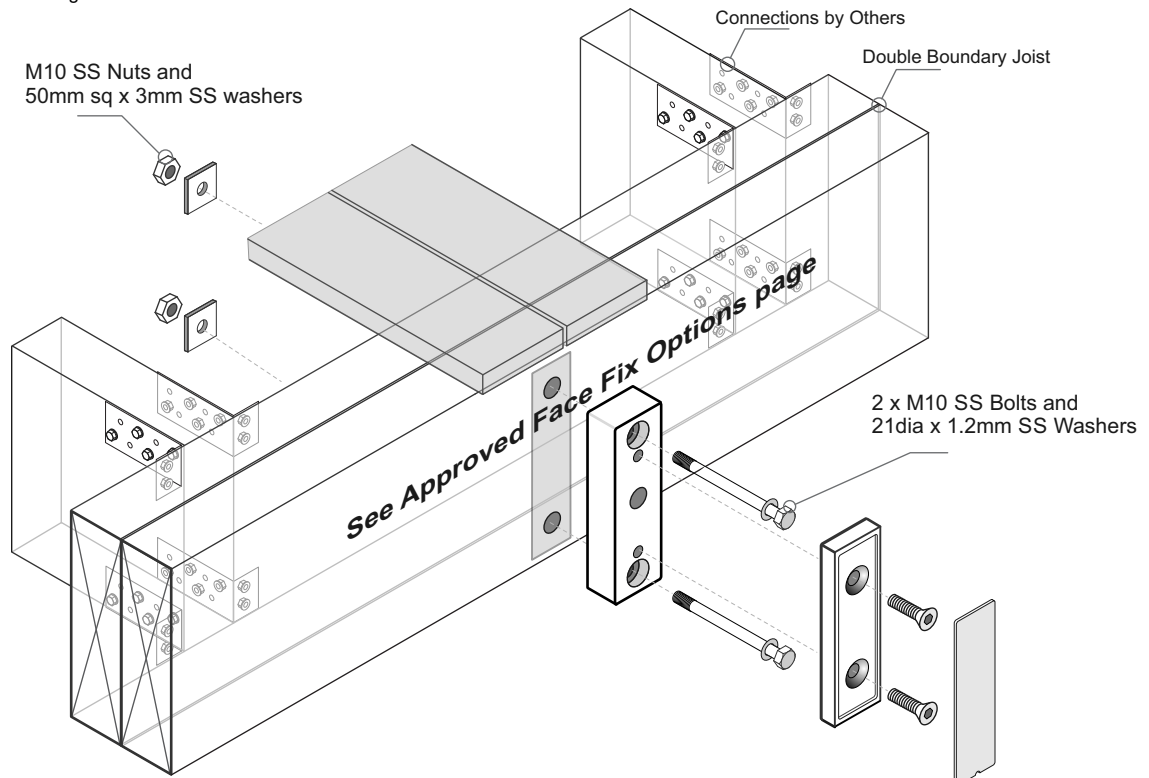
- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



See Approved Face Fix Options page

Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - All fixings must be Stainless Steel
- 4 - Max 50mm spacing back to structure including JVB125 Spacer



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing
Complies with NZS3604:2011 - Double Boundary Joists

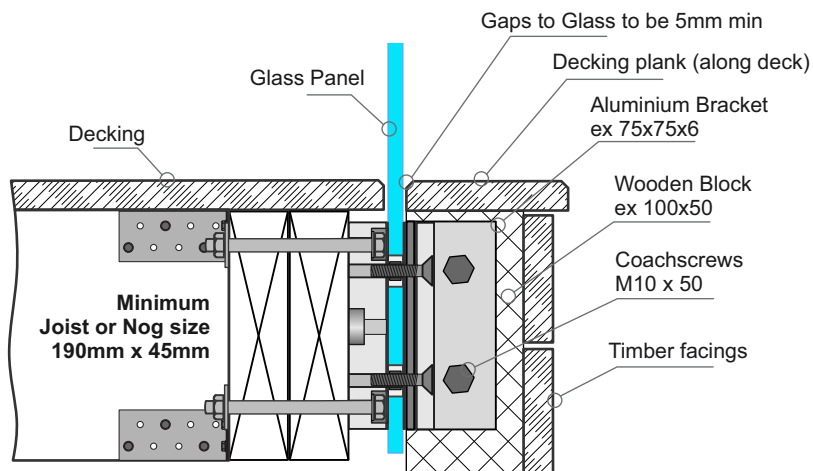
Typical HIDDEN FACE Fix to Timber - M10 SS Bolts

Occupancy AA Other and C3 - Residential				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

Note: Decking Planks and Timber Facings must be screw attached to allow for removal under B2 Durability requirements

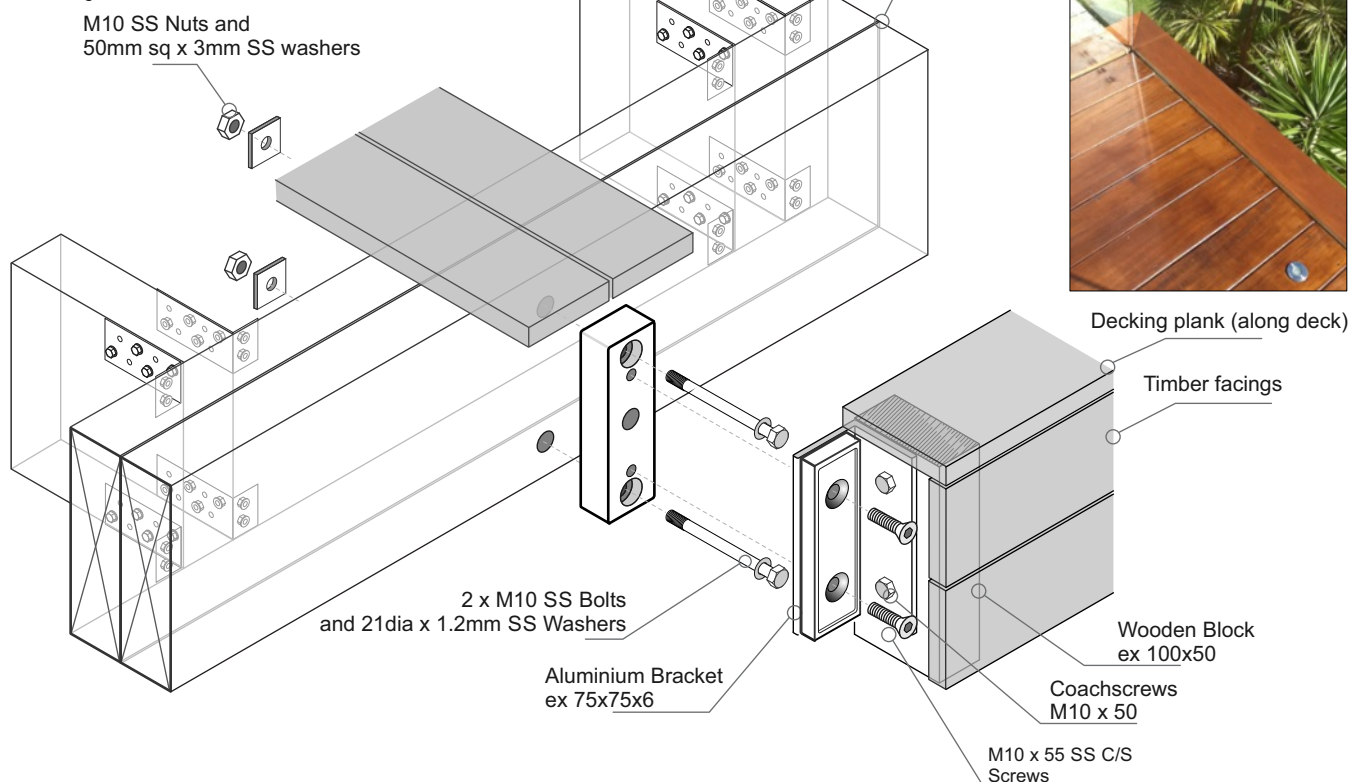


Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - All fixings must be Stainless Steel
- 4 - Max 60mm spacing back to structure including JVB125 Spacer

General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing

Typical FACE Fix to Steel - M12 SS Bolt

Occupancy A, A Other and C3 - Residential

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial

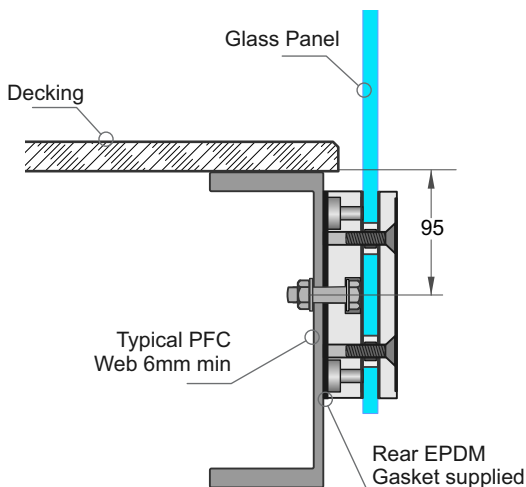
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

General Notes:

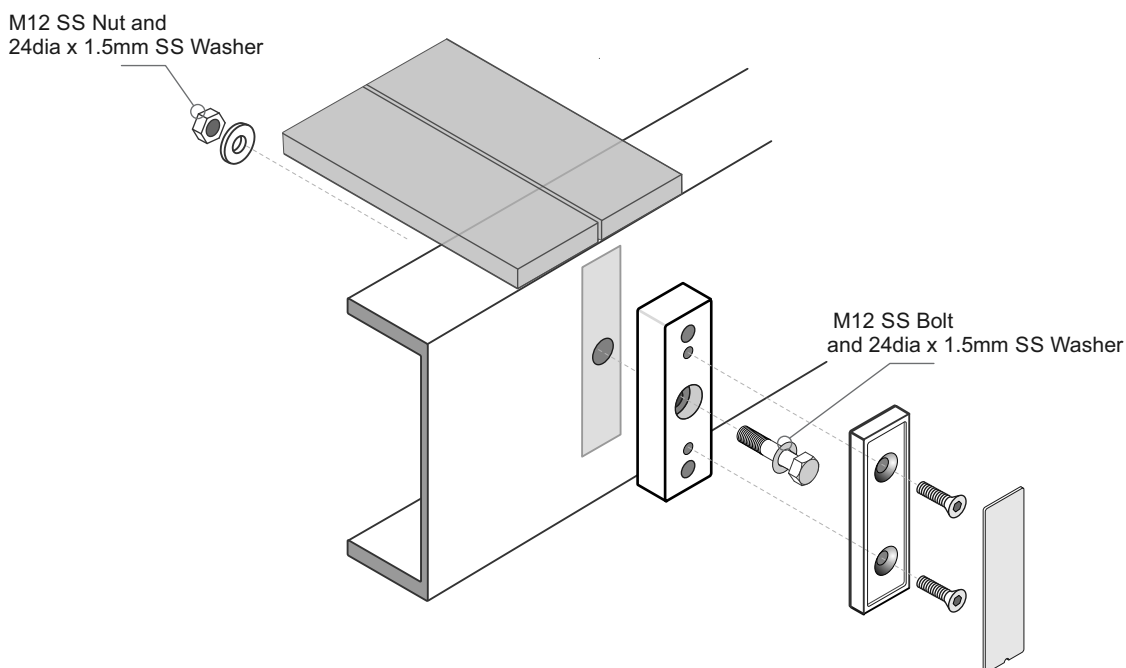
- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 10kN



Important Installation Notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
- 3 - The Rear EPDM Gasket supplied must be installed between the Clamp and Steel
- 4 - All fixings must be Stainless Steel
- 5 - Max 60mm spacing back to structure including JVB125 Spacer

M12 SS Nut and
24dia x 1.5mm SS Washer



Typical FACE Fix to Steel - M10 SS Bolts

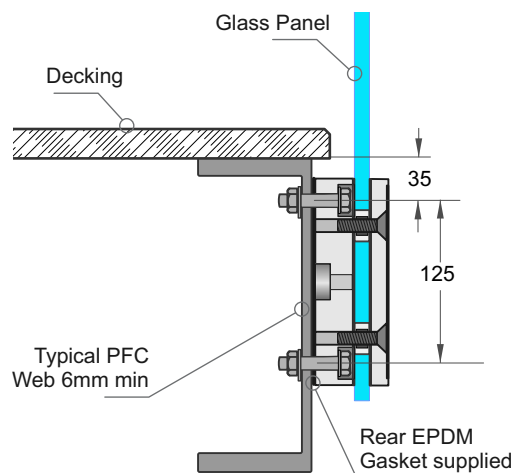
Occupancy A, A Other and C3 - Residential				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

General Notes:

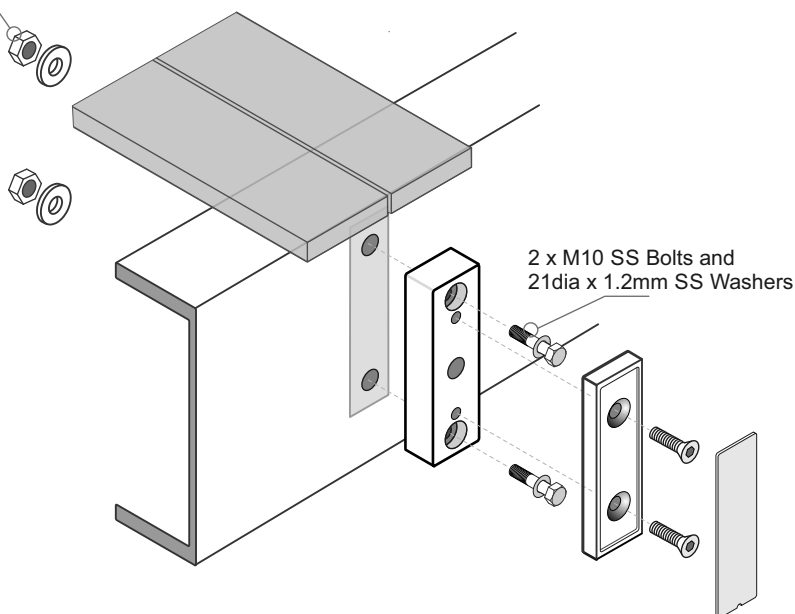
- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - The Rear EPDM Gasket supplied must be between the Clamp and Steel
- 3 - All fixings must be Stainless Steel
- 4 - Max 60mm spacing back to structure including JVB125 Spacer

M12 SS Nut and
24dia x 1.5mm SS Washer



JURALCO

www.juralco.co.nz ph (09) 478 8018

Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing

Typical FACE Fix to Steel, Wooden Packers - M10 SS Bolts

Occupancy A, A Other and C3 - Residential

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial

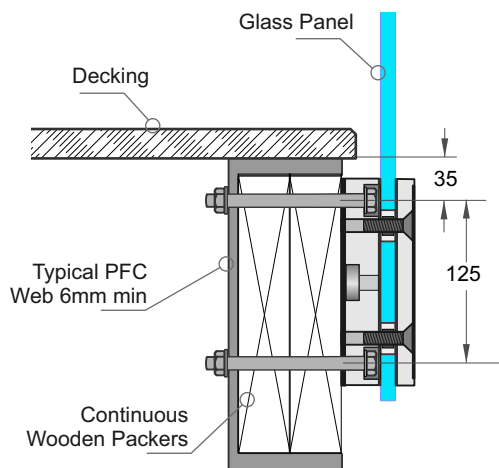
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

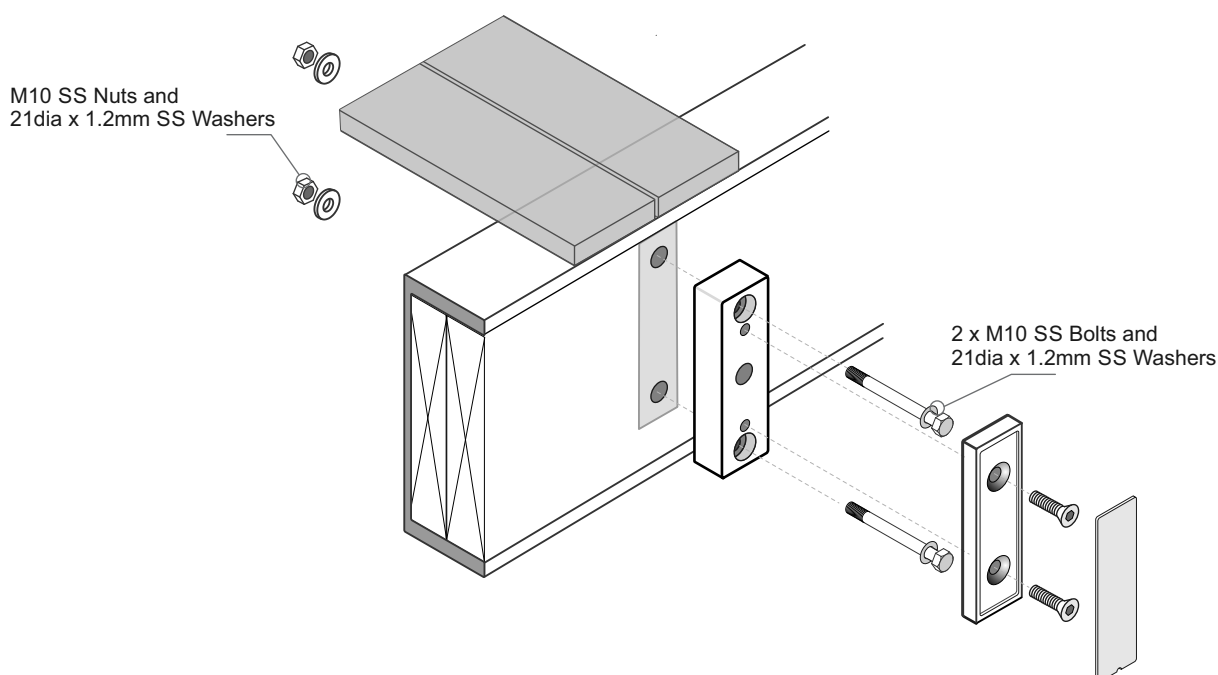
General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - All fixings must be Stainless Steel
- 4 - Max 60mm spacing back to structure including JVB125 Spacer



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing

Typical FACE Fix to Steel, Wooden Packers - M10 SS Bolts

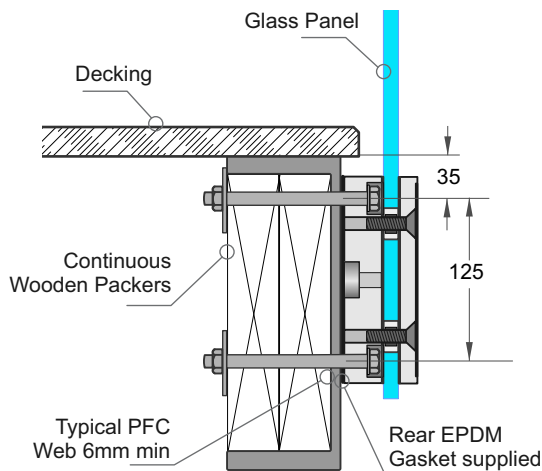
Occupancy A, A Other and C3 - Residential				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		

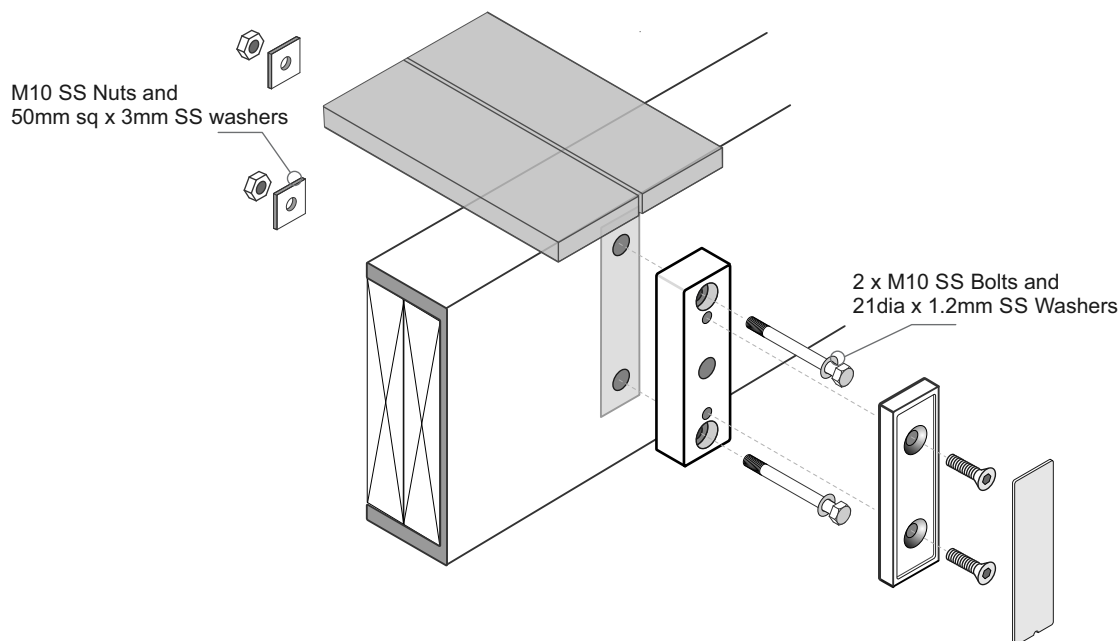
General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)
- 4 - Max Tension Load per Face fixing for this mounting = 7kN



Important Installation notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
Timber SG8 minimum strength
- 3 - The Rear EPDM Gasket supplied must be between the Clamp and Steel
- 4 - All fixings must be Stainless Steel
- 5 - Max 60mm spacing back to structure including JVB125 Spacer



JURALCO

www.juralco.co.nz ph (09) 478 8018

Juralco Edgetec® JH Clamp Balustrade System

Issue 1-26 v1

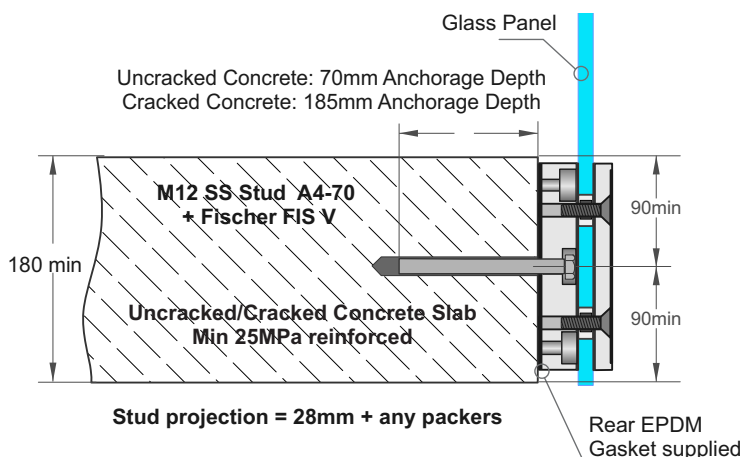
Page 21

Typical FACE Fix to Concrete - M12 SS Stud

Occupancy A, A Other and C3 - Residential				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only				
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450		
	High	1400		
	Very High	1350		
	Extra High	1275		



Important Installation Notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
- 3 - The Rear EPDM Gasket supplied must be between the Clamp and Concrete
- 4 - Use Loctite on the Nut
- 5 - All fixings must be Stainless Steel
- 6 - Max 60mm spacing back to structure including JVB125 Spacer

General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)

Fischer Injection System FIS V



Injection Resin FIS V 410 C
or FIS V 300 T

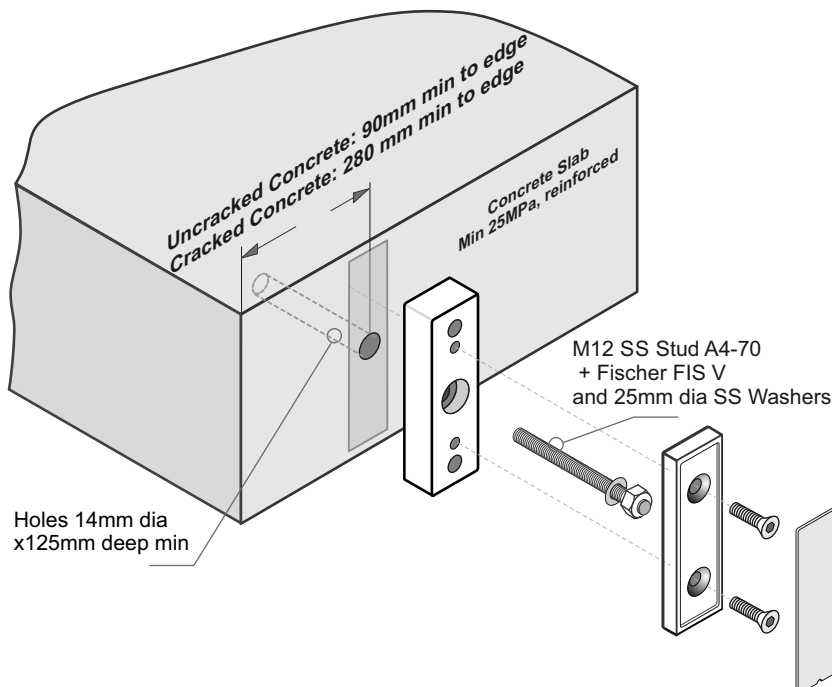
Installation details

Stud M12 SS, A4-70
Drill hole diameter 14mm
Drill hole depth 125mm
Anchorage depth 70mm for Cracked,
185mm for Uncracked

Drilling method - hammer drilling

4 times blowing,
4 times brushing,
4 times blowing

Volume of resin per drill hole
8 ml/4 scale division



Juralco Edgetec® JH Clamp Balustrade System - Typical Fixing

Typical FACE Fix to Concrete - M10 SS Studs

Occupancy A, A Other and C3 - Residential

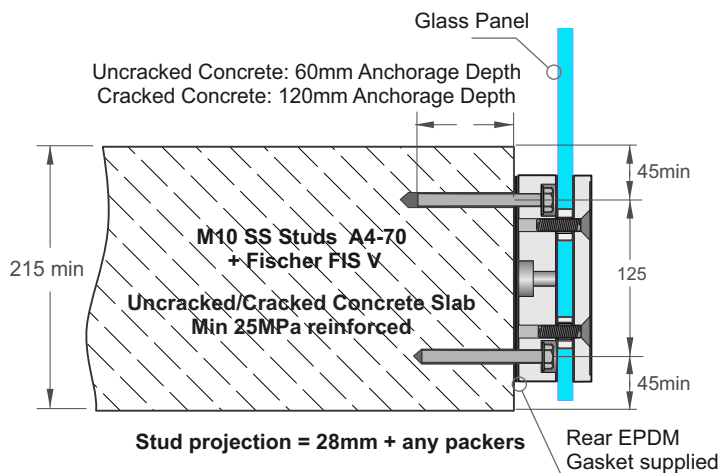
Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang (max)
12T, 13.52SG, 15.2L	Extra High	1050	400	200
	Very High	1150	500	
12T	Very High	1250	400	

Occupancy B, E and C3 - Commercial

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
17.2L	Very High	1250	400	200
17.52SG	Extra High	1250	500	
15T	Extra High	1350	400	

Pool Fence Only

Glass Type	Wind Zone	Balustrade Height (max)	Clamp Spacing (max)	Glass Overhang
12T, 13.52SG	Medium	1400	500	200
	High	1350		
	Very High	1275		
15T, 17.52SG	Medium	1450	500	200
	High	1400		
	Very High	1350		
	Extra High	1275		



Important Installation Notes: (applies to all 3 x types)

- 1 - The Project Engineer must ensure the structure can support the appropriate horizontal and vertical loads
- 2 - Substructure shown indicatively only
- 3 - The Rear EPDM Gasket supplied must be between the Clamp and Concrete
- 4 - Use Loctite on the Nut
- 5 - All fixings must be Stainless Steel
- 6 - Max 60mm spacing back to structure including JVB125 Spacer

General Notes:

- 1 - Glass thickness, mm
Glass type T = Toughened, L = Laminated
SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to typical Layouts for Max Panel widths and the use of Top Interlinking Rails (T and L) or Stiffener Brackets (L and SG)

Fischer Injection System FIS V



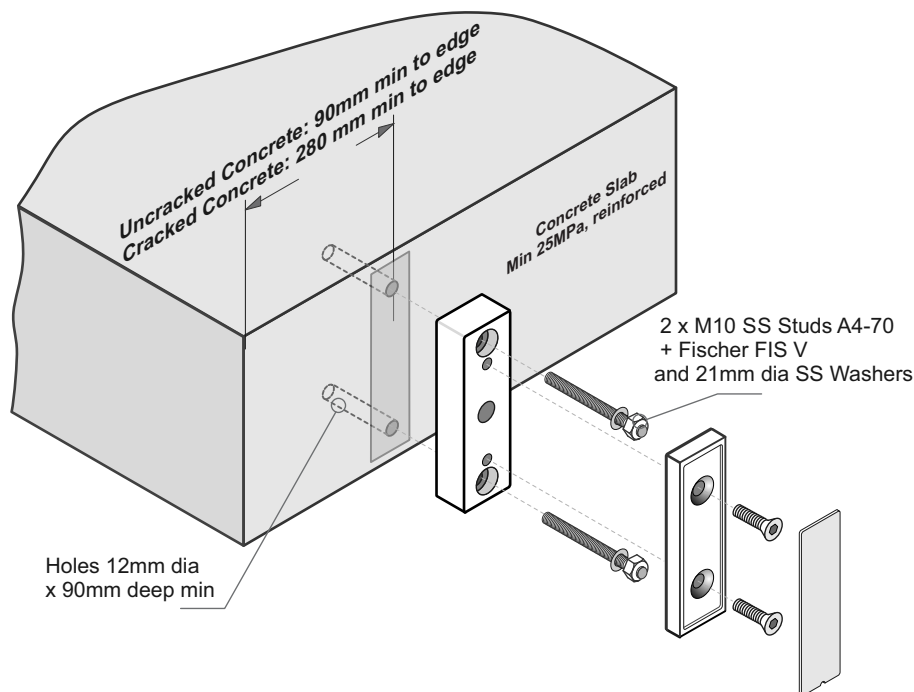
Injection Resin FIS V 410 C
or FIS V 300 T

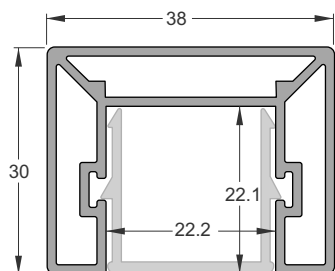
Installation details

Stud M10 SS, A4-70
Drill hole diameter 12mm
Drill hole depth 90mm
Anchorage depth 60mm for Un-cracked,
120mm for Cracked

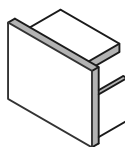
Drilling method - hammer drilling
4 times blowing,
4 times brushing,
4 times blowing

Volume of resin per drill hole
8 ml/4 scale division

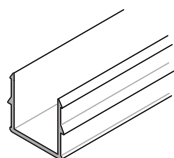




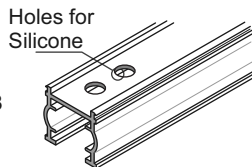
Rectangular Interlinking Top Rail
Part No JET/220/5.8
Also showing Infill Clip, for use in between Glass Panels



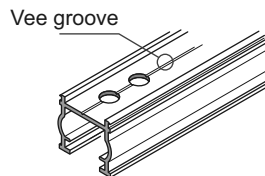
Interlinking Top Rail End Cap
Part No JET 37



Infill Clip
Part No JET/215/5.8



Interlinking Top Rail Gasket for 12 mm Toughened Glass
Part No JET/Gasket 12/2.9



Interlinking Top Rail Gasket for 15 mm Toughened Glass
Part No JET /Gasket 15/2.9

1 - 12, 15mm Glass and Gasket

Application Notes:

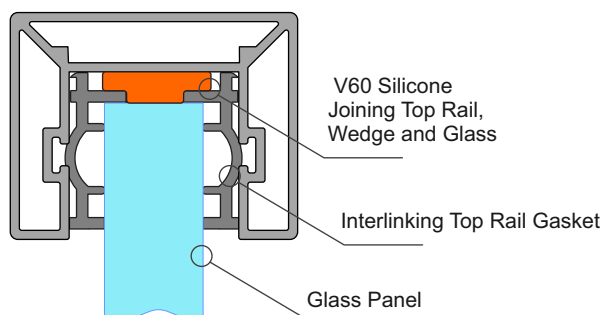
- Cut short lengths of Gasket (50mm) and place say every 700mm.
- Cut/adjust Interlinking rail to correct dimensions, test in place.
- Remove all, install full cut lengths of Gasket to glass top edge

- Assemble Top Rail + Joiners and suitable End plates

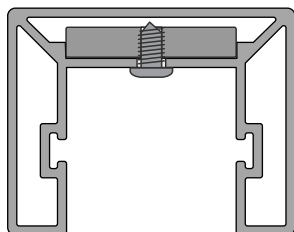
- Place blobs of V60 silicone in every Gasket hole
- Then place Top Rail extrusion + Joiners and End plates in place clipping firmly to Gasket
- Tape all down, wait 24 hrs to fully bond. Clean up.

Note: Ends must be attached to structure or post,
- Joins must have a suitable joiner plate

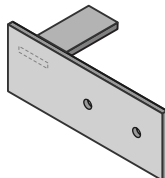
12mm Glass and Gasket shown



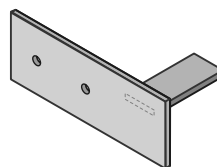
2 - End Plate Brackets



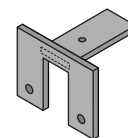
End Plate
Tabs all 22.5 x 4mm SS.



Interlinking Top Rail Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail Wall type End Plate
SS. 120x45mm
JET 40RH

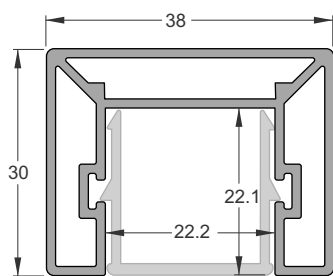


Interlinking Top Rail End Bracket
SS. 60mm x 46mm
JET 42

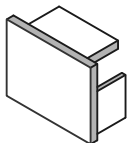
End Plates: (After cutting extrusions to length)

- With End Plate in place, spot drill from below for position
- Drill out to SS tab to 3mm dia, extrusion to 4mm dia
- Use No 6 x 1/4in SS ST Pan sq drive Screw, 2 per plate.
- End Plate must be securely attached to Post or structure.

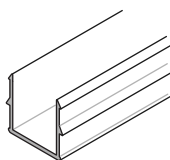
Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets



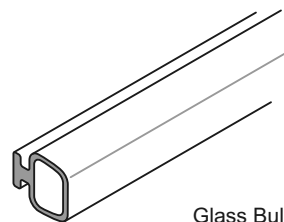
Rectangular Interlinking Top Rail
JET/220/5.8
Also showing Infill Clip, for
use in between Glass Panels



Interlinking Top
Rail End Cap
JET 37



Infill Clip
Part No JET/215/5.8



Glass Bulb Seal
Part No JET/IB/CVRBLB250

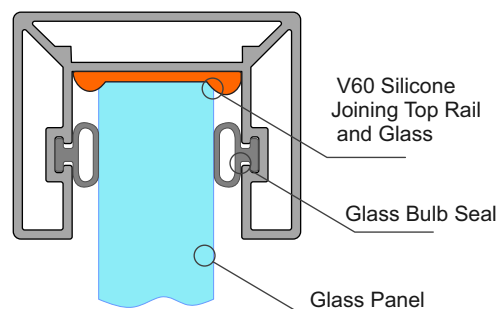
1 - 15.2, 17.2mm Glass and Gasket

Application Notes:

- Assemble Top Rail + Joiners and suitable End plates
- Place Full lengths of Bulb seal in place.
- Place blobs of V60 silicone along top edge of Glass at similar spacings to Gasket on previous page.
- Then place Top Rail extrusion and bulb seals firmly onto Glass.
- Tape all down, wait 24 hrs to fully bond. Clean up.

Note: Ends must be attached to structure or post,
- Joins must have a suitable joiner plate

15.2mm Glass and Gasket shown

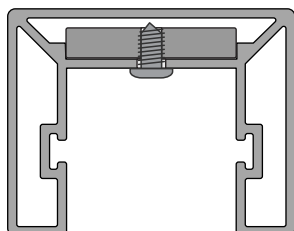


V60 Silicone
Joining Top Rail
and Glass

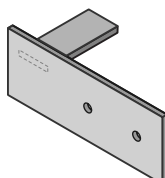
Glass Bulb Seal

Glass Panel

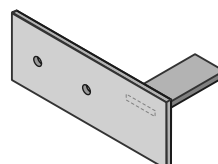
2 - End Plate Brackets



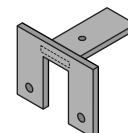
End Plate
Tabs all 22.5 x 4mm SS.



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH



Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
JET 42

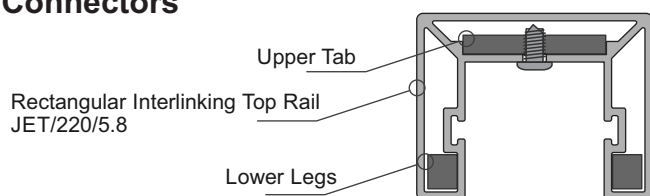
End Plates: (After cutting extrusions to length)

- With End Plate in place, spot drill from below for position
- Drill out to SS tab to 3mm dia, extrusion to 4mm dia
- Use No 6 x 1/4in SS ST Pan sq drive Screw, 2 per plate.
- End Plate must be securely attached to Post or structure.

Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets

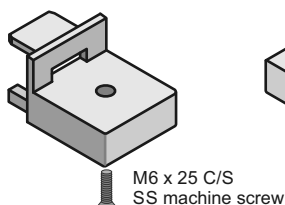
Juralco 38mm Rectangular Interlinking Top Rail - Corner Connectors and Joiners

1 - Connectors

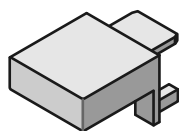


- Swivel Kits: (After cutting extrusions to length)
- With Swivel in place, spot drill from below for position
 - Drill out Swivel to 3mm dia, extrusion to 4mm dia
 - Use No6 x 1/4in SS ST Pan sq drive screws, 2 x ea side of joint
 - Both sides must be attached.
 - Join together with the M6 x 25 C/S SS Screw

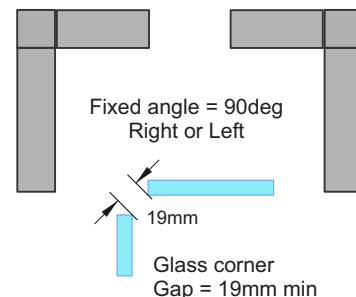
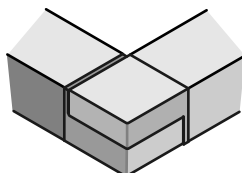
Interlinking Top Rail
Horizontal Fixed
90 deg Connector
JET 45A



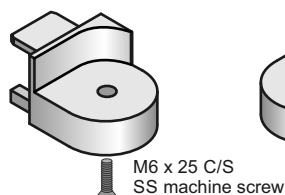
Interlinking Top Rail
Horizontal Fixed
90 deg Connector
JET 45B



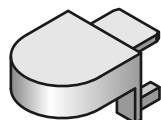
Rectangular Interlinking Top Rail
Horizontal Fixed 90deg Kit
JET220/90deg Corner Kit
(JET 45A and B + screw)



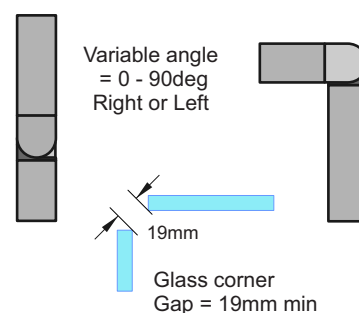
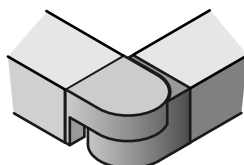
Interlinking Top Rail
Horizontal 0 - 90deg
Swivel Connector
JET 46A



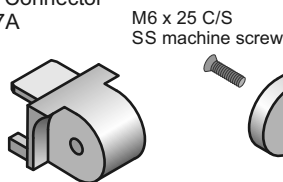
Interlinking Top Rail
Horizontal 0 - 90deg
Swivel Connector
JET 46B



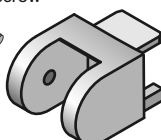
Rectangular Interlinking Top Rail
Horizontal 0 - 90 deg Right/Left
Swivel Connector Kit
JET220/Horizontal Adj Corner Kit
(JET 46A and B + screw)



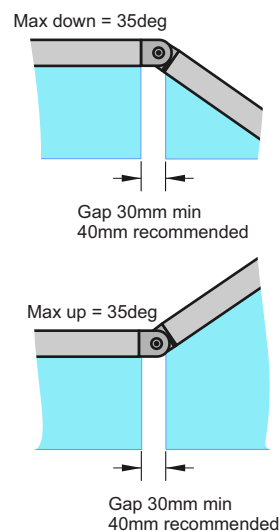
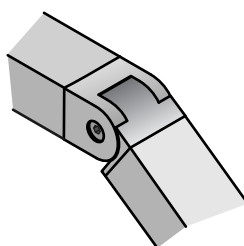
Interlinking Top Rail
Vertical 35deg up
to 35 deg down
Swivel Connector
JET 47A



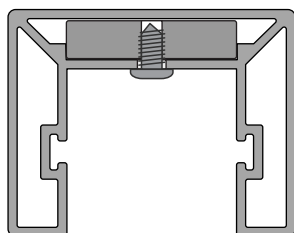
Interlinking Top Rail
Vertical 35deg up
to 35 deg down
Swivel Connector
JET 47B



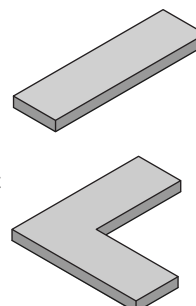
Interlinking Top Rail Vertical 35deg up
to 35 deg down. Swivel Connector Kit
JET220/Vertical Adj Corner Kit
(JET 47A and B + screw)



2 - Joiners



- Joiners: (After cutting extrusions to length)
- With Joiner in place, spot drill from below for position
 - Drill out to joiner to 3mm dia, extrusion to 4mm dia
 - Use No 6 x 1/4in SS ST Pan sq drive screws, 2 x ea side of joint
 - Both ends must be attached.
 - Joins, where required must be at the end of Glass Panels



Interlinking Top Rail
Straight Joiner
80x22.8x5mm
JET 30

Interlinking Top Rail
Corner Joiner
75x75x5mm
JET 31

Joiners both 22.5 x 5mm Aluminium



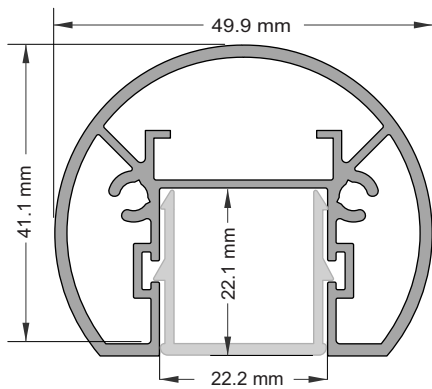
JURALCO

www.juralco.co.nz ph (09) 478 8018

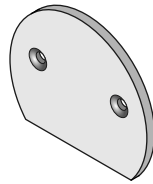
Juralco Edgetec® JH Clamp Balustrade System

Issue 1-26 v1

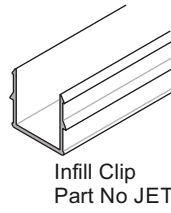
Page 26



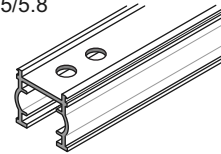
Round Interlinking Top Rail
Part No JET/211/5.8
Also showing Infill Clip, for
use in between Glass Panels



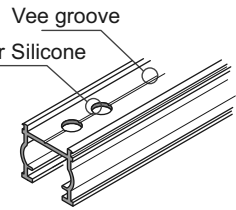
Round Interlinking
Top Rail End Cap
Part No JET/231



Infill Clip
Part No JET/215/5.8



Interlinking Top Rail Gasket
for 12 mm Toughened Glass
Part No JET /Gasket 12/2.9



Interlinking Top Rail Gasket
for 15 mm Toughened Glass
Part No JET /Gasket 15/2.9

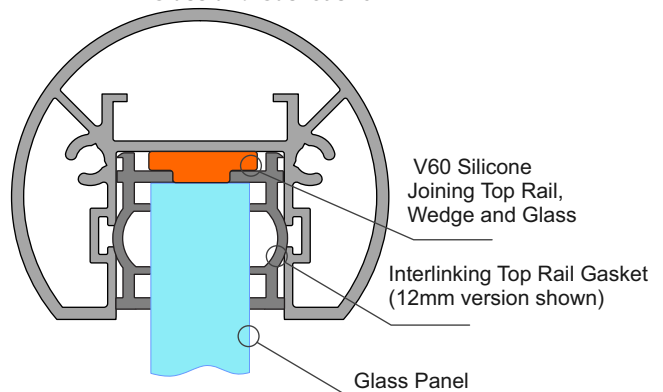
1 - 12, 15mm Glass and Gasket

Application Notes:

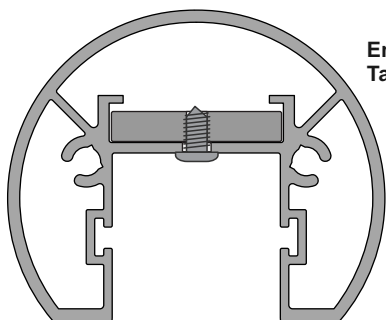
- Cut short lengths of Gasket (50mm) and place say every 700mm.
- Cut/adjust Interlinking rail to correct dimensions, test in place.
- Remove all, install full cut lengths of Gasket to glass top edge
- Assemble Top Rail + Joiners and suitable End plates
- Place blobs of V60 silicone in every Gasket hole
- Then place Top Rail extrusion + Joiners and End plates in place clipping firmly to Gasket
- Tape all down, wait 24 hrs to fully bond. Clean up.

Note: Ends must be attached to structure or post,
- Joins must have a suitable joiner plate

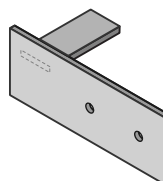
12mm Glass and Gasket shown



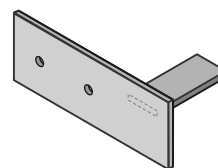
2 - End Plate Brackets



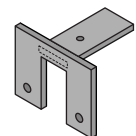
End Plate
Tabs all 22.5 x 4mm SS.



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH

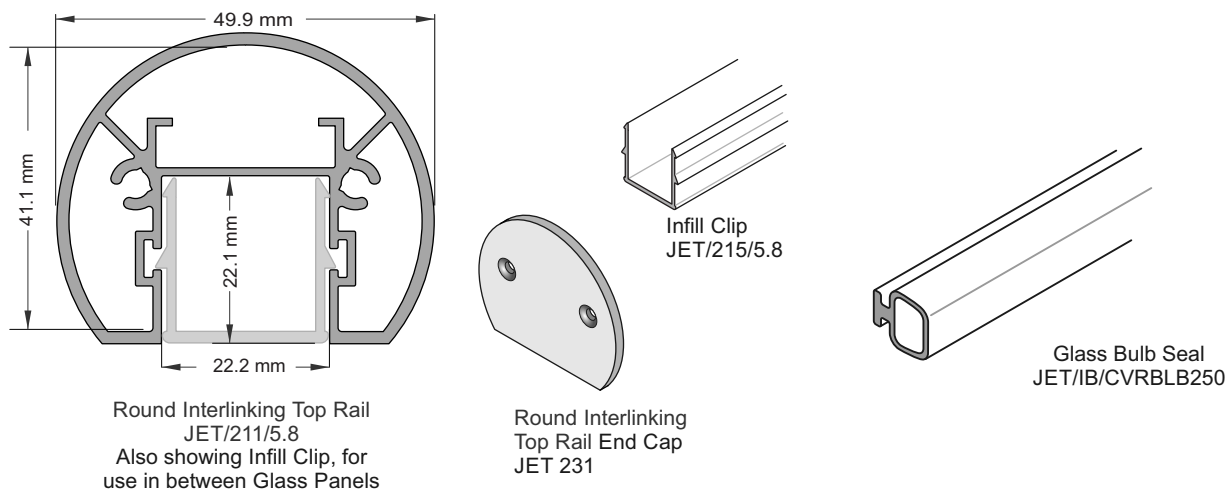


Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
JET 42

End Plates: (After cutting extrusions to length)

- With End Plate in place, spot drill from below for position
- Drill out to SS tab to 3mm dia, extrusion to 4mm dia
- Use No 6 x 1/4in SS ST Pan sq drive Screw, 2 per plate
- End Plate must be securely attached to Post or structure.

Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets



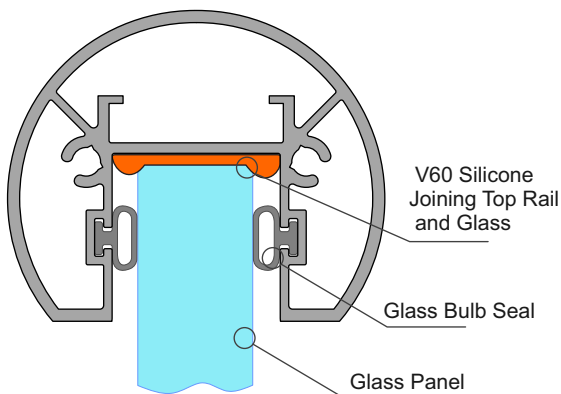
1 - 15.2, 17.2mm Glass and Gasket

15.2mm Glass and Gasket shown

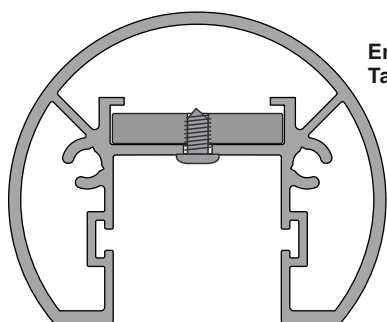
Application Notes:

- Assemble Top Rail + Joiners and suitable End plates
- Place Full lengths of Bulb seal in place.
- Place blobs of V60 silicone along top edge of Glass at similar spacings to Gasket on previous page.
- Then place Top Rail extrusion and bulb seals firmly onto Glass.
- Tape all down, wait 24 hrs to fully bond. Clean up.

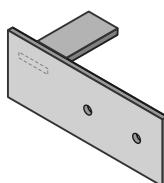
Note: Ends must be attached to structure or post,
- Joins must have a suitable joiner plate



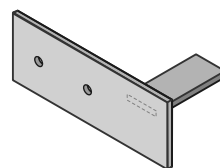
2 - End Plate Brackets



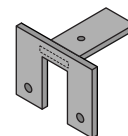
End Plate
Tabs all 22.5 x 4mm SS.



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH

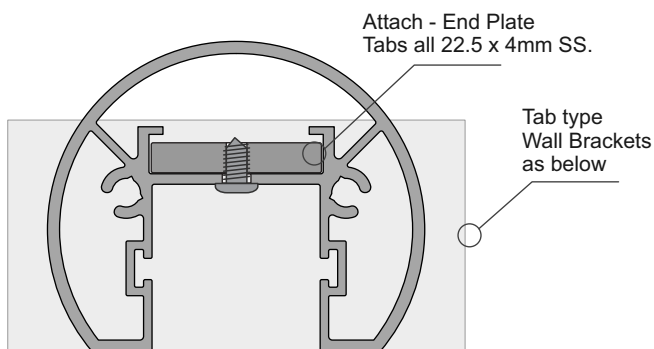


Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
JET 42

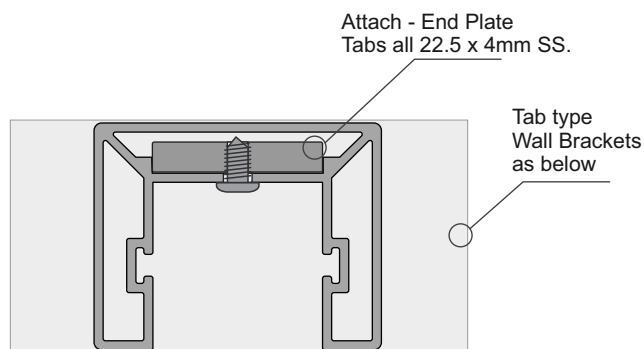
End Plates: (After cutting extrusions to length)

- With End Plate in place, spot drill from below for position
- Drill out to SS tab to 3mm dia, extrusion to 4mm dia
- Use No 6 x 1/4in SS ST Pan sq drive Screw, 2 per plate
- End Plate must be securely attached to Post or structure.

Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets

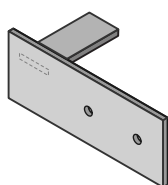


Round Interlinking Top Rail Extrusion
JET/211/5.8

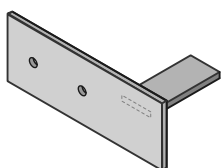


Interlinking Top Rail Extrusion
JET/220/5.8

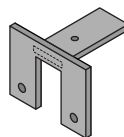
Interlinking Top Rail End Bracket Options - Both types above - Tab attach Type



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



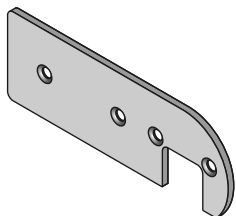
Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH



Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
JET 42

**Tabs all 22.5 x 4mm SS.
Front faces all 3mm SS**

Interlinking Top Rail End Bracket Options - Round Type only - Attach into Screw ports



Interlinking Top Rail
Wall type offset End Plate
Round Rail type only
120x42x3mm, Al
C/s both sides = RH or LH
JET 233



Interlinking Top Rail
Wall type End Plate
Round Rail type only
50x58x5mm, Al
JET 232

General Notes:

- All fixings to be Stainless Steel - PVC Tape layer between Structure and Bracket
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down and in tension

Note : Fixing to Steel

- use 2 off 8g SS TEK Screws or M6 SS Bolts
- Steel 2mm min thickness
- Steel 300MPa minimum
- 15mm min distance to any Edges

Note : Fixing to Timber Wall

- use 2 off 8g SS Screws, 35mm min into studs.
- use Sika Supergrip 30mins
- 30mm min distance to Horizontal Edge
- If Weatherboard use suitable predrilled Wedge
- Timber stud wall to be designed and detailed in accordance with NZS 1720.1:2022 Timber Structures Part 1 - Design methods or NZ3604

Note : Fixing to Juralco EDGE Post

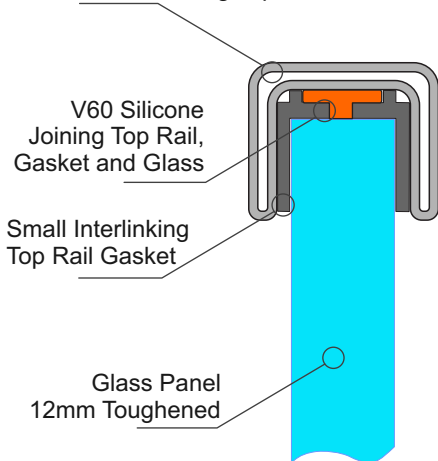
- use 2 off 8g x 25 SS PK Screws

Note : Fixing to Concrete Wall

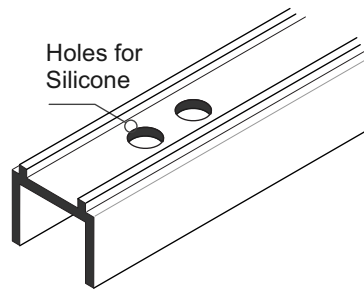
- use 2 off M6 x70 SS Screw Anchors
- Solid Concrete min 20Mpa
- Block wall Concrete filled/Reinforced
- 140mm min Wall thickness
- 70mm min distance to Horizontal Edge
- 100mm min distance to Vertical Edge
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229

Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets

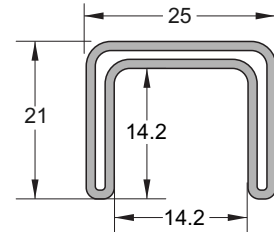
Small SS Interlinking Top Rail



25mm SS Interlinking Top Rail



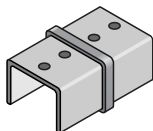
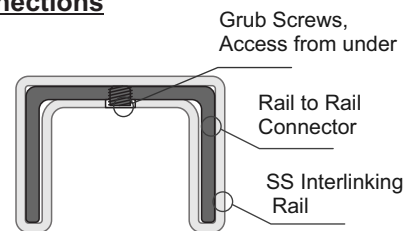
SMALL SS INTERLINKING
TOP RAIL GASKET
JET/490GT/12/2.9 (Black)



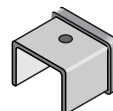
SMALL SS INTERLINKING TOP RAIL
JET/490/5.8/SSS JET/490/5.8/SCC
Duplex 2205

25mm SS Interlinking Rail Connections

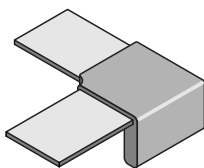
Note : All these Brackets use M5 x 6 SS Grub Screws.
If necessary these holes must be Drilled + tapped M5, as shown.
The under side of the Interlinking Rail must be drilled
M6 to match M5 tapped holes positions, for access to Grub screws
- Joins, where required must be at the end of Glass Panels
Available as Satin(SSS) or Powdercoated SCC finishes



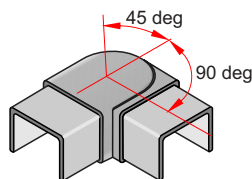
180deg INLINE JOINER
Duplex 2205
JET491/SSS JET491/SCC
21mm x 25mm x 51mm deep



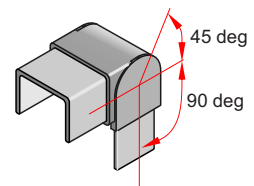
END CAP Duplex 2205
JET492/SSS JET492/SCC
21mm x 25mm x 25mm deep



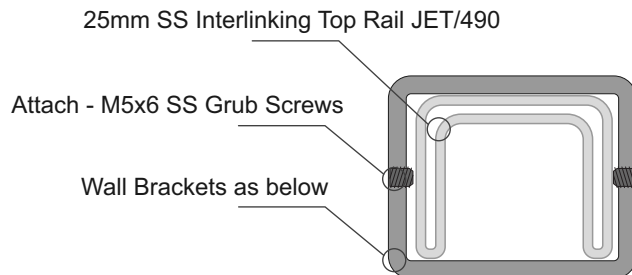
90deg JOINER Duplex 2205
JET493/SSS JET493/SCC
21mm x 80mm x 80mm



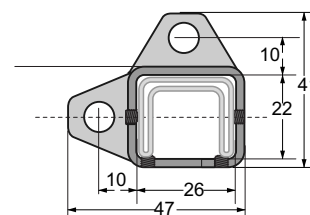
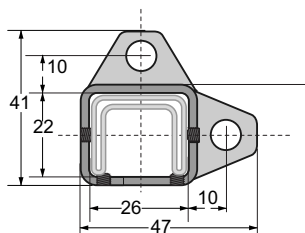
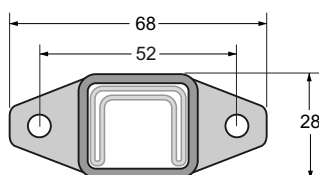
+90 to - 45 deg ADJUSTABLE HORIZONTAL
JOINER Duplex 2205
JET494/SSS JET494/SCC
21mm x 25mm x 75mm overall deep



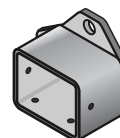
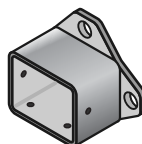
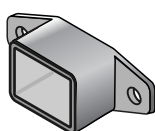
+90 to - 45 deg ADJUSTABLE
VERTICAL JOINER Duplex 2205
JET495/SSS JET495/SCC
21mm x 25mm x 73mm overall deep



Brackets for Fixing to Wall or End Post for 25mm SS Interlinking Rail



Note : All these Brackets use M5x6mm SS Grub Screws



WALL BRACKET Duplex 2205
JET496/SSS JET/496/SCC
68mm x 28mm x 30mm deep

WALL BRACKET - RH. Duplex 2205
JET497/RH/SSS JET497/RH/SCC
41mm x 47mm x 30mm deep

WALL BRACKET - LH Duplex 2205
JET497/LH/SSS JET497/RH/SCC
41mm x 47mm x 30mm deep

General Notes:

- All fixings to be Stainless Steel. - PVC Tape layer between Structure and Bracket
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down and in tension

Note : Fixing to Steel

- use 2 off 8g SS TEK Screws or M6 SS Bolts
- Steel 2mm min thickness
- Steel 300MPa minimum
- 15mm min distance to any Edges

Note : Fixing to Timber Wall

- use 2 off 8g SS Screws, 35mm min into studs.
- use Sika Supergrip 30mins
- 30mm min distance to Horizontal Edge
- If Weatherboard use suitable predrilled Wedge
- Timber stud wall to be designed and detailed in accordance with NZS 1720.1:2022 Timber Structures Part 1 - Design methods or NZ3604

Note : Fixing to Juralco EDGE Post

- use 2 off 8g x 25 SS PK Screws

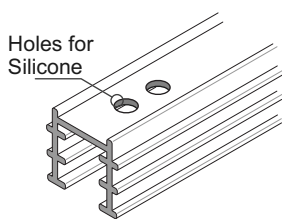
Note : Fixing to Concrete Wall

- use 2 off M6 x70 SS Screw Anchors
- Solid Concrete min 20Mpa
- Block wall Concrete filled/Reinforced
- 140mm min Wall thickness
- 70mm min distance to Horizontal Edge
- 100mm min distance to Vertical Edge
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229

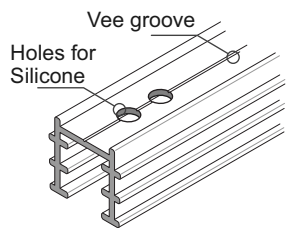
Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets

Juralco 40mm SS Interlinking Top Rail

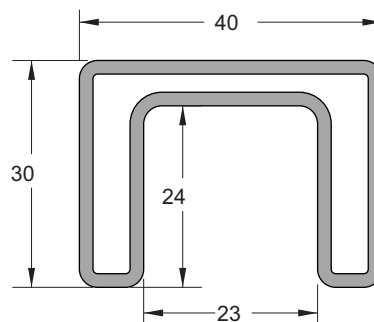
This page applies to 12mm and 15mm Toughened Glass and 15.2mm and 17.2mm Laminated Glass only



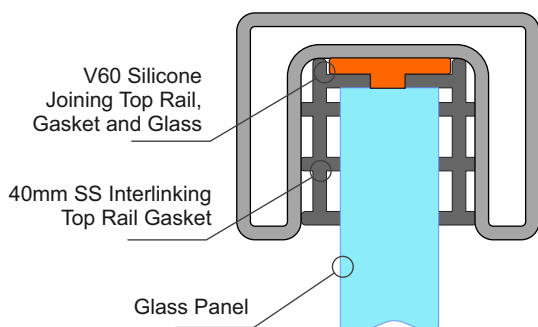
SS Interlinking Top Rail
12mm Glass Gasket
JET/430GT/12/2.9



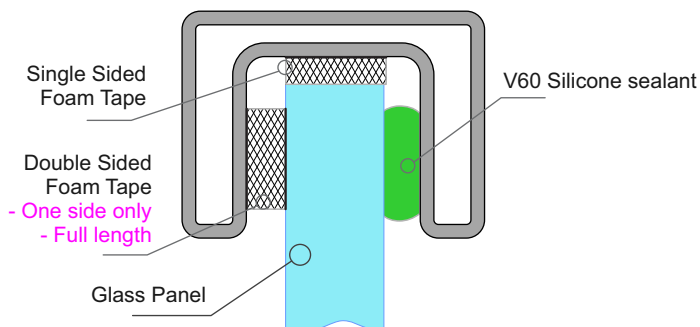
SS Interlinking Top Rail
15mm Glass Gasket
JET/430GT/15/2.9



SS INTERLINKING TOP RAIL
JET/430/PSS/5.8



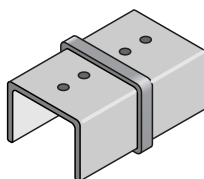
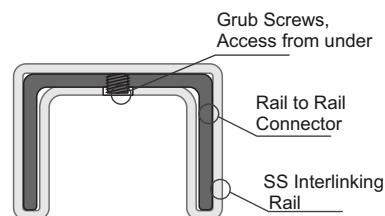
**Use Gasket for 12mm and
15mm Toughened Glass**



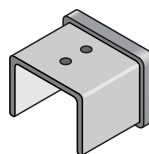
**Use Foam Tape for 15.2mm and
17.2mm Laminated Glass**

40mm SS Interlinking Rail Connectors

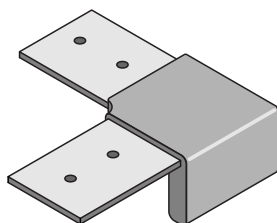
Note : All these Brackets use M5 x 6 SS Grub Screws.
If necessary these holes must be Drilled + tapped M5, as shown.
The under side of the Interlinking Rail must be drilled
M6/7 to match M5 tapped holes positions, for access to Grub screws
- Joins, where required must be at the end of Glass Panels



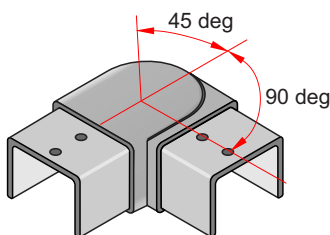
180deg INLINE JOINER 2205
JET/431/PSS
60mm x 40mm x 30mm deep



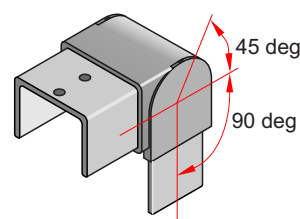
END CAP 2205
JET/432/PSS
33mm x 40mm x 30mm deep



90deg JOINER 2205
JET/433/PSS
95mm x 95mm x 30mm deep

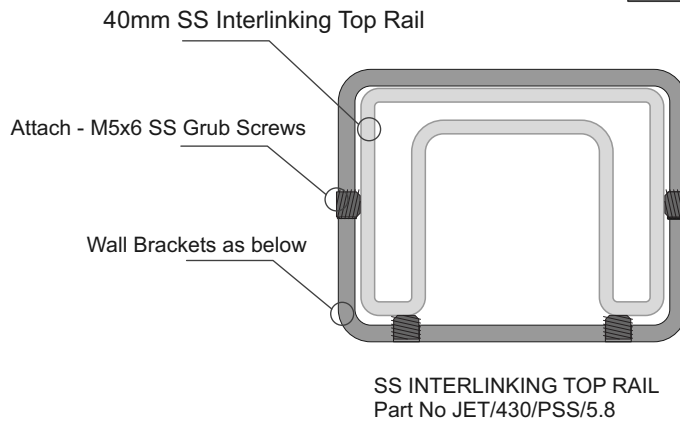


+90 to - 45 deg ADJUSTABLE
HORIZONTAL JOINER 2205
JET/434/PSS
70mm x 70mm x 30mm deep



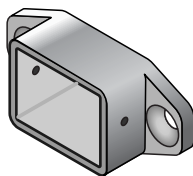
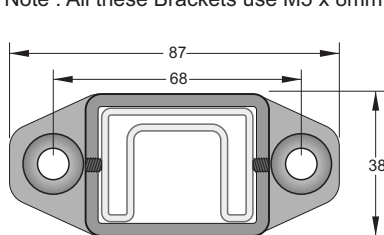
+90 to - 45 deg ADJUSTABLE
VERTICAL JOINER 2205
JET/435/PSS
60mm x 60mm x 40mm wide



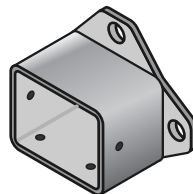
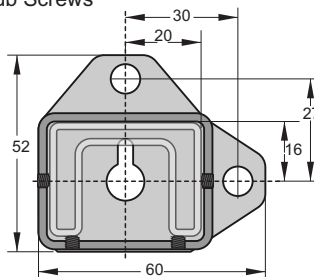


Brackets for Fixing to Wall or End Post for 40mm SS Interlinking Rail

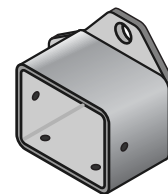
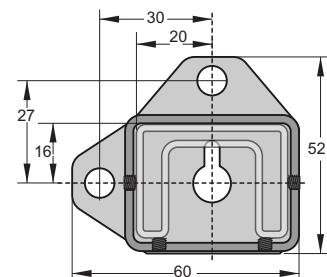
Note : All these Brackets use M5 x 8mm SS Grub Screws



WALL BRACKET 2 FIX 2205
Part No JET/436/PSS
87mm x 37mm x 25mm deep



WALL BRACKET 2 FIX - RH 2205
Part No JET/437/RH/PSS
52mm x 60mm x 33mm deep



WALL BRACKET 2 FIX - LH 2205
Part No JET/437/LH/PSS
52mm x 60mm x 33mm deep

General Notes:

- All fixings to be Stainless Steel - PVC Tape layer between Structure and Bracket
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down and in tension

Note : Fixing to Steel

- use 2 off 8g SS TEK Screws or M6 SS Bolts
- Steel 2mm min thickness
- Steel 300MPa minimum
- 15mm min distance to any Edges

Note : Fixing to Timber Wall

- use 2 off 8g SS Screws, 35mm min into studs.
- use Sika Supergrip 30mins
- 30mm min distance to Horizontal Edge
- If Weatherboard use suitable predrilled Wedge
- Timber stud wall to be designed and detailed in accordance with NZS 1720.1:2022 Timber Structures Part 1 - Design methods or NZ3604

Note : Fixing to Juralco EDGE Post

- use 2 off 8g x 25 SS PK Screws

Note : Fixing to Concrete Wall

- use 2 off M6 x70 SS Screw Anchors
- Solid Concrete min 20Mpa
- Block wall Concrete filled/Reinforced
- 140mm min Wall thickness
- 70mm min distance to Horizontal Edge
- 100mm min distance to Vertical Edge
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229

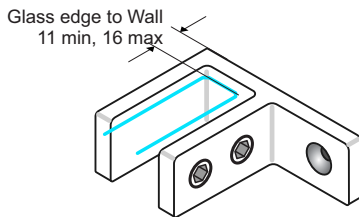
Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets

Juralco Glass Panel Stiffener Brackets

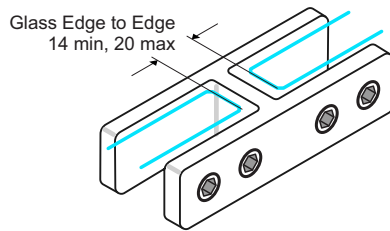
Top Edge, Frameless
Glass Stiffeners.

- Install 200mm max from Glass Top edge
- Supplied as a kit, with screws, a variety of Gaskets and a SS clamp Plate
- Duplex 2205 SS construction.
Polished (PSS), Satin (SSS)
or Powder coat SCC Finishes

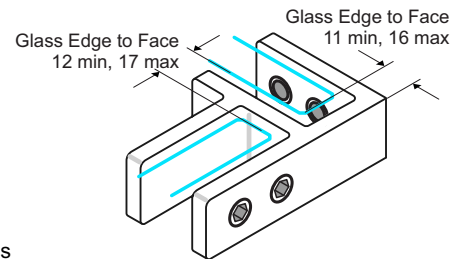
**Applies to 15.2mm Toughened Laminated Glass and 15.52mm SentryGlas.
Also for 12mm Toughened Glass for Pool Fences only**



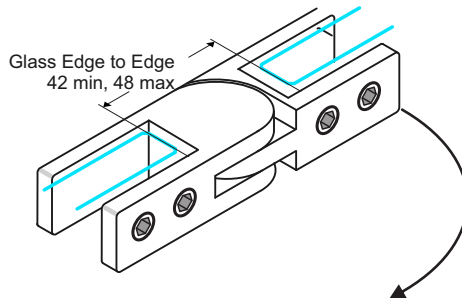
90 Deg Glass to Wall
75x505x25mm
Part No JET/72/PSS
Part No JET/72/SSS
Part No JET/72/SCC



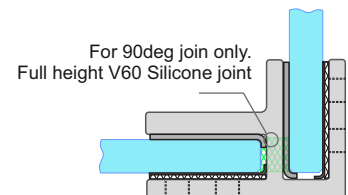
180 Deg Glass to Glass
70x34x25mm
Part No JET/71/PSS
Part No JET/71/SSS
Part No JET/71/SCC



90 deg Glass to Glass
65x55x25mm
Part No JET/70/PSS
Part No JET/70/SSS
Part No JET/70/SCC



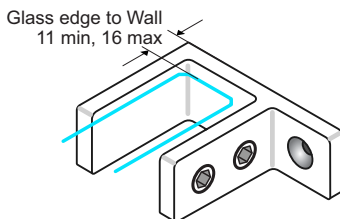
90 - 180 Deg
Adjustable Glass to Glass
135x34x25mm
Part No JET/73/PSS
Part No JET/73/SSS
Part No JET/73/SCC



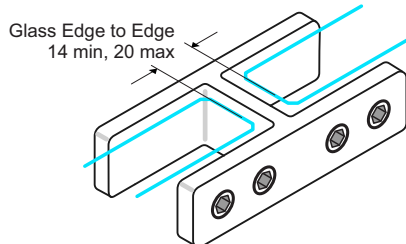
Top Edge, Frameless
Glass Stiffeners.

- Install 200mm max from Glass Top edge
- Supplied as a kit, with screws, a variety of Gaskets and a SS clamp Plate
- Duplex 2205 SS construction.
Polished (PSS), Satin (SSS)
or Powder coat SCC Finishes

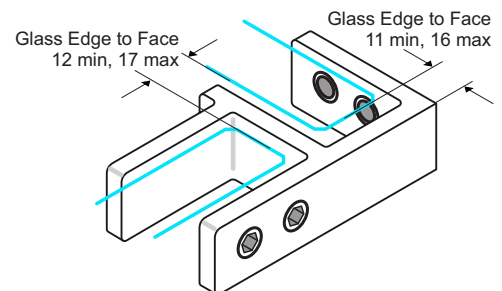
**Applies to 17.2mm Toughened Laminated Glass and 17.52mm SentryGlas only.
Up to max Barrier height 1250mm**



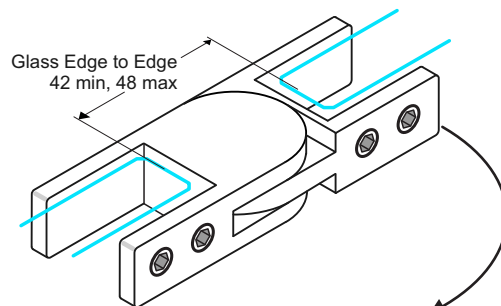
90 Deg Glass to Wall
65x55x25mm
Part No JET/82/PSS
Part No JET/82/SSS
Part No JET/82/SCC



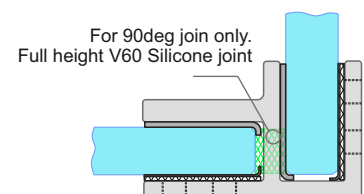
180 Deg Glass to Glass
103x39x25mm
Part No JET/81/PSS
Part No JET/81/SSS
Part No JET/81/SCC



90 deg Glass to Glass
88x55x25mm
Part No JET/80/PSS
Part No JET/80/SSS
Part No JET/80/SCC

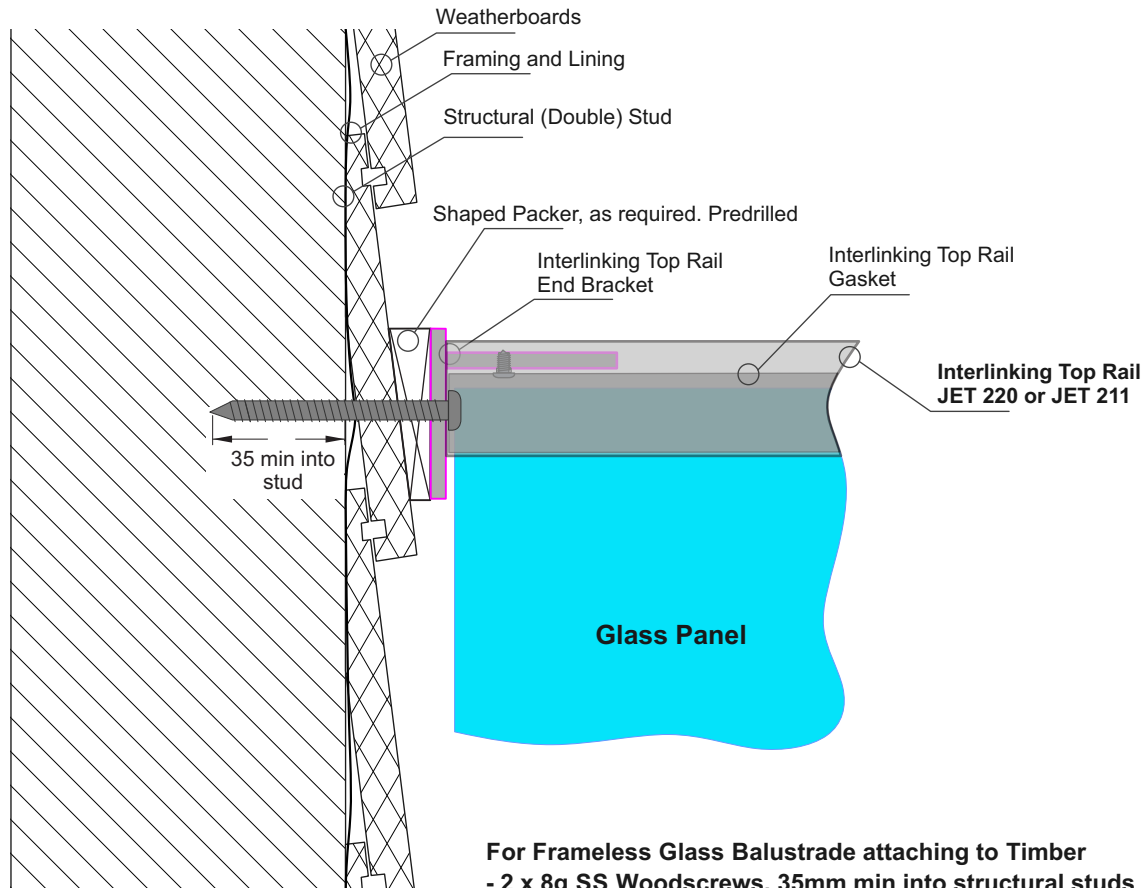


90 - 180 Deg Adjustable
Glass to Glass
145x39x25mm
Part No JET/83/PSS
Part No JET/83/SSS
Part No JET/83/SCC



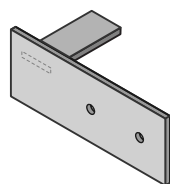
Juralco Interlinking Top Rail End Bracket fixing to a Timber Wall

**Applies to 38mm Rectangular and
50mm Round Interlinking Top Rails only**

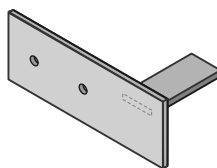


For Frameless Glass Balustrade attaching to Timber
- 2 x 8g SS Woodscrews, 35mm min into structural studs
- 20mm min to edge of stud in all directions

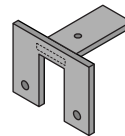
Interlinking Top Rail End Bracket Options - Drawing above shows JET40



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH



Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
Part No JET 42

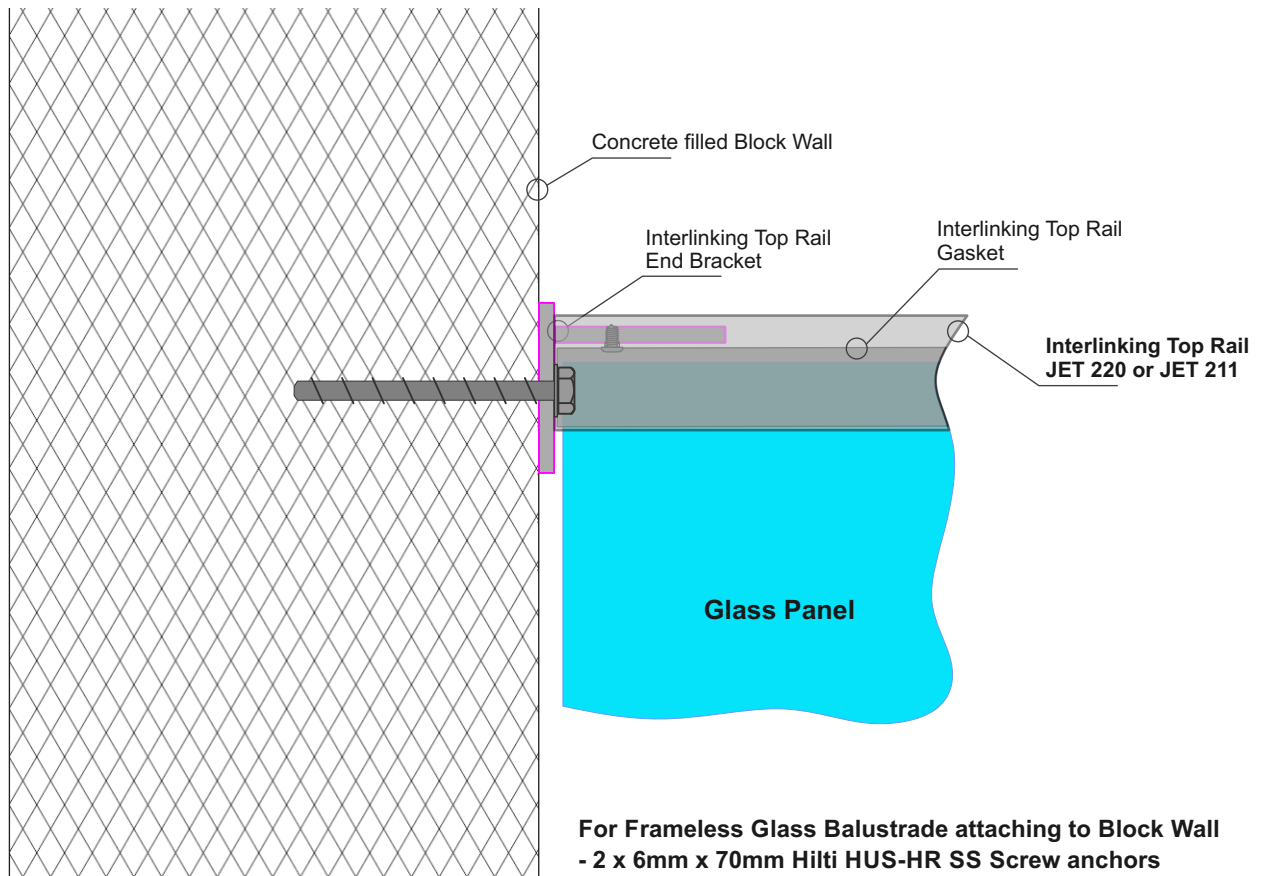
Notes:

- All fixings to be stainless steel
- Timber stud wall to be designed by Project structural engineer for loads imposed by Balustrade.
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down.
- Minimum Stud size = 90mm x 45mm
- Minimum Timber grade = SG8
- Timber stud wall to be designed and detailed in accordance with NZS 1720.1:2022 Timber Structures Part 1 - Design methods or NZ3604



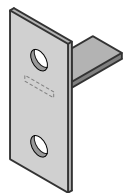
**Juralco Interlinking Top Rail End Bracket
fixing to a Concrete Wall**

**Applies to 38mm Rectangular and
50mm Round Interlinking Top Rails only**

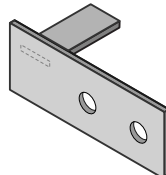


For Frameless Glass Balustrade attaching to Block Wall
 - 2 x 6mm x 70mm Hilti HUS-HR SS Screw anchors
 - For concrete drill 6mmØ holes
 - 70mm min to side edge of concrete, 100mm to top edge.

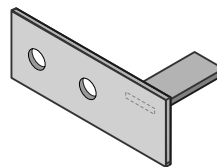
Interlinking Top Rail End Bracket Options - Drawing above shows JET40



Interlinking Top Rail
Wall type End Plate
SS. 100x45mm
JET 38



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
JET 40RH

Drill out holes to 9mmØ

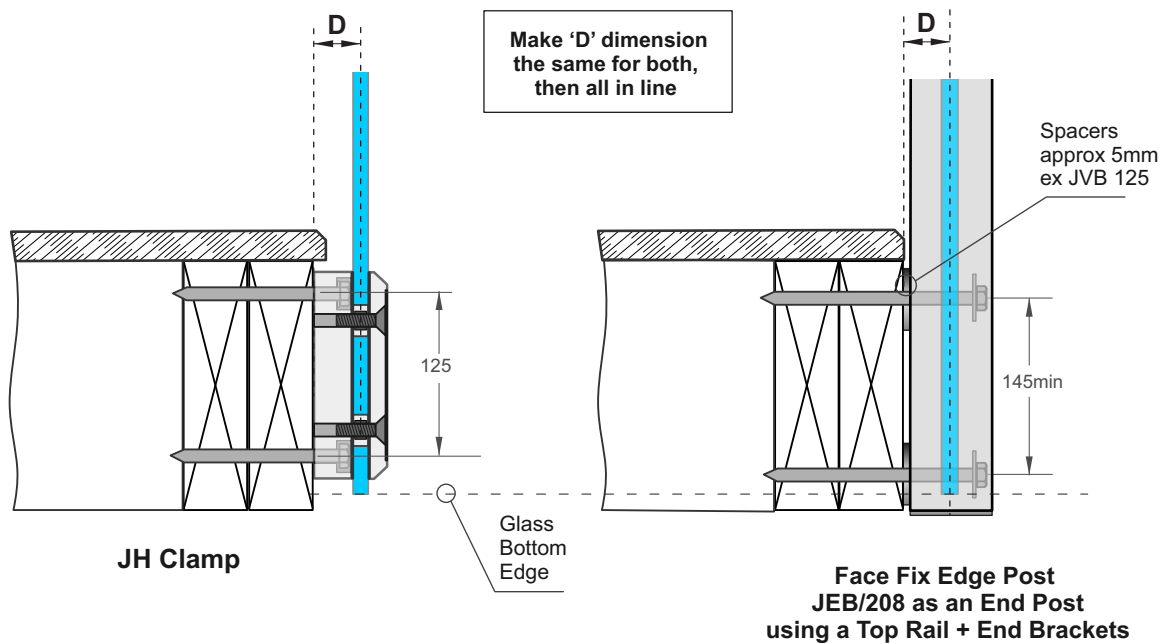
Notes:

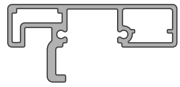
- All fixings to be stainless steel
- Blockwall to be designed by Project structural engineer for loads imposed by Balustrade.
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down.
- Minimum blockwork thickness = 140mm
- Minimum core fill concrete strength = 17.5MPa
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229



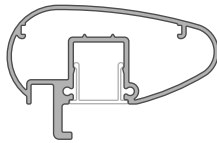
Juralco Edgetec® JH Clamp Anchor Balustrade System
Interlinking Top Rail for attaching to a Edge balustrade End Post
where Wall fixing not suitable

**Applies Interlinking Top Rails suitable for
 12mm, 15mm Toughened Glass and
 15.2mm or 17.2mm Laminated Glass,**

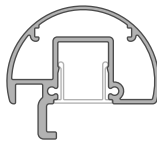




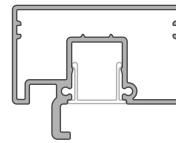
INTERLINKING RAIL
JEB/222/5.8



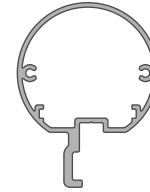
AEROFOIL HANDRAIL
JEB/217/5.8



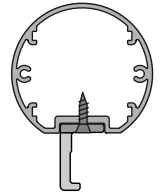
ROUND HANDRAIL
JEB/209/5.8



RECTANGULAR
HANDRAIL
JEB/216/5.8



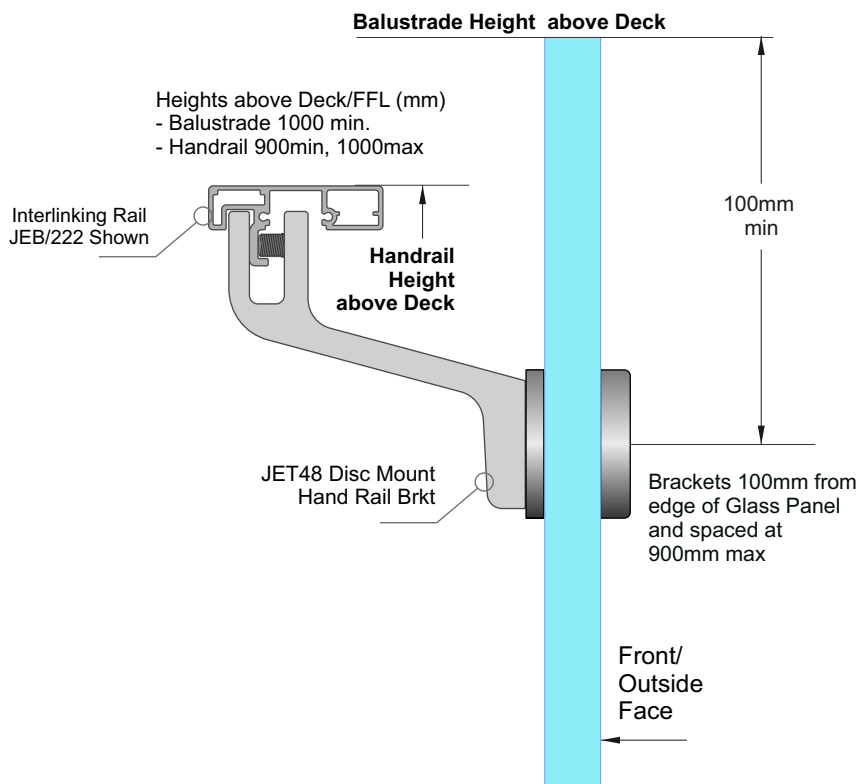
CIRCULAR
HANDRAIL
JEB/221/5.8



CIRCULAR
HANDRAIL
JEB/223/5.8
+ Clip JEC38

Suitable Interlinking Rail and Handrails (as Interlinking Rails)

Interlinking or Handrails on Deck side.



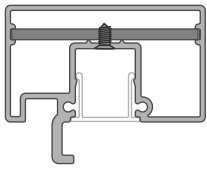
Frameless Glass Systems

Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets. Applies to Handrails used as Interlinking Rails

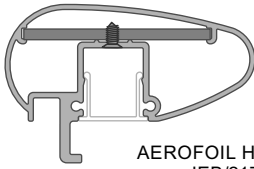
Juralco Handrail Components - Joiners

Rectangular Handrails and
75mm Aerofoil - End Cap,
Straight and 90deg corners

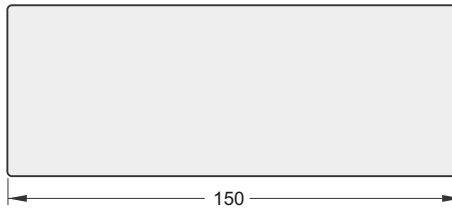
All ex 3mm Aluminium



RECTANGULAR
HANDRAIL
JEB/216/5.8

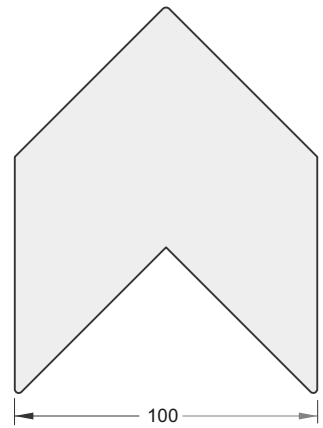


AEROFOIL HANDRAIL
JEB/217/5.8



Rectangular and 75mm Aerofoil Inline Joiner
Use 56.5 x 3 flat bar JA/189/5.0

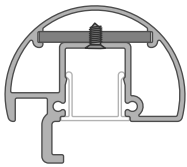
Use No6 x 1/4in SS pan sq drive screws, 2 ea side of joint



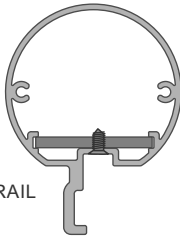
Rectangular and 75mm Aerofoil
90deg Corner Joiner JEC 01

Round and Circular
Handrail, End Cap,
Straight and 90deg corners

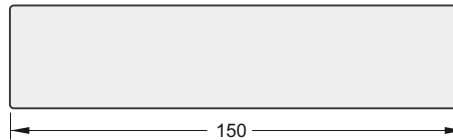
All ex 3mm Aluminium



ROUND HANDRAIL
JEB/209/5.8

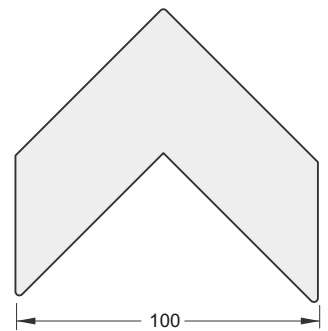


CIRCULAR HANDRAIL
JEB/221/5.8



Round and Circular Inline Joiner
Use 34.0 x 3 flat bar JA/188/5.0

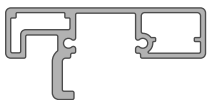
Use No6 x 1/4in SS pan sq drive screws, 2 ea side of joint



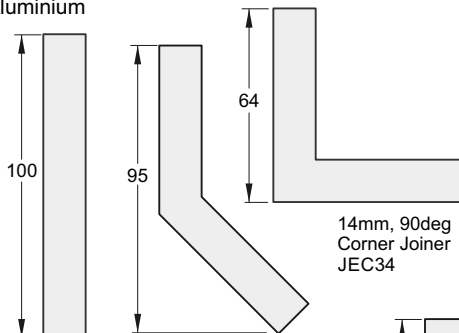
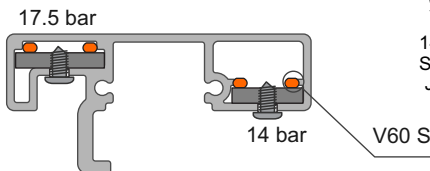
Round and Circular 90deg Corner Joiner
JEC 04

Interlinking Rail
End Cap, Straight
135 deg and 90deg corners

All ex 3mm Aluminium



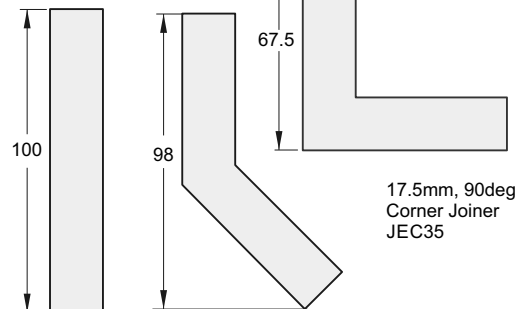
STANDARD HANDRAIL
JEB/222/5.8



14mm
Straight Joiner
JEC32

14mm, 135deg
Corner Joiner
JEC36

14mm, 90deg
Corner Joiner
JEC34



17.5mm
Straight Joiner
JEC33

17.5mm, 135deg
Corner Joiner
JEC37

17.5mm, 90deg
Corner Joiner
JEC35

Joiners:

- With Joiner in place, spot drill from below for position
- Drill out joiner to 3mm dia, extrusion to 4mm dia
- Use No 6 x 1/4in SS ST Pan sq drive screw
- Insert dobs of V60 Silicone inside cavities before inserting
- Both ends to be attached.
- Joins must be within 300mm of Post
- Minimum distance between screw and end of handrail is 10mm



JURALCO

www.juralco.co.nz ph (09) 478 8018

Juralco Edgetec® JH Clamp Balustrade System

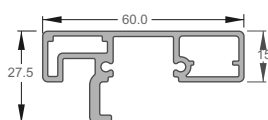
Issue 1-26 v1

Page 39

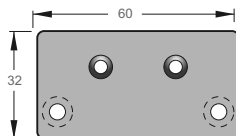
Juralco Handrail End Plates for Attaching to a Structure or Edge Deck mounted Post

End Caps for Handrails, Wall or Edge Post attach for JEB 222, 217, 209, 216 and 221 Handrails

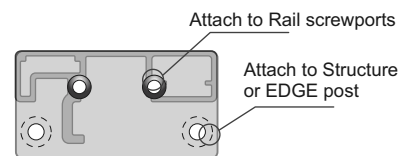
End Caps
all ex 3mm Aluminium



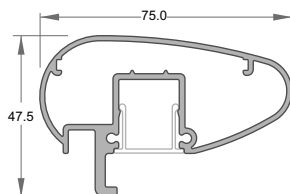
HANDRAIL
Part No JEB/222/5.8



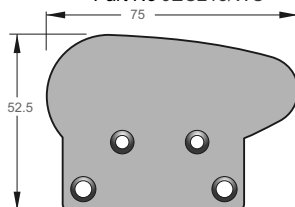
HANDRAIL
WALL ATTACH END PLATE
Part No JEC215/WC



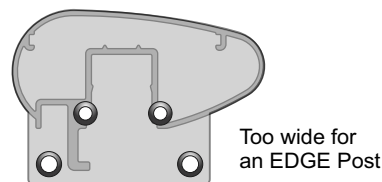
For RH and LH



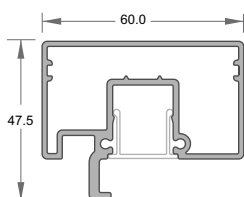
AEROFOIL HANDRAIL
Part No JEB/217/5.8



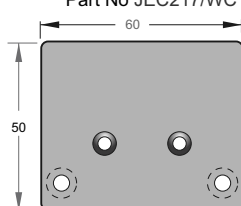
HANDRAIL
WALL ATTACH END PLATE
Part No JEC217/WC



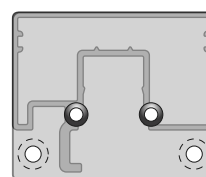
For RH and LH



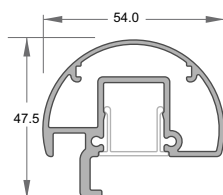
RECTANGULAR
HANDRAIL
Part No JEB/216/5.8



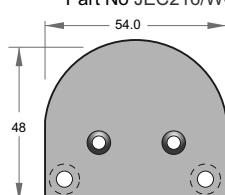
HANDRAIL
WALL ATTACH END PLATE
Part No JEC216/WC



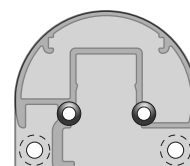
For RH and LH



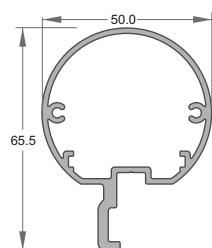
HALF ROUND HANDRAIL
Part No JEB/209/5.8



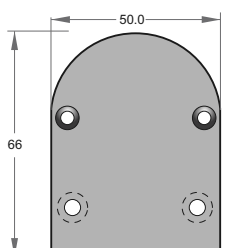
HANDRAIL
WALL ATTACH END PLATE
Part No JEC209/WC



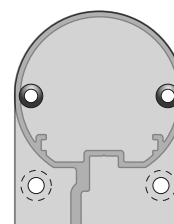
For RH and LH



CIRCULAR HANDRAIL
Part No JEB/221/5.8



HANDRAIL
WALL ATTACH END PLATE
Part No JEC221/WC



For RH and LH

General Notes: - All fixings to be Stainless Steel. - EPDM layer between Structure and End Cap
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down and in tension

Note : Fixing to Steel

- use 2 off 8g SS TEK Screws or M6 SS Bolts
- Steel 2mm min thickness
- Steel 300MPa minimum
- 15mm min distance to any Edges

Note : Fixing to Juralco EDGE Post

- use 2 off 8g x 25 SS PK Screws

Note : Fixing to Timber Wall

- use 2 off 8g SS Screws, 35mm min into studs.
- use Sika Supergrip 30mins
- 30mm min distance to Horizontal Edge
- If Weatherboard use suitable predrilled Wedge
- Timber stud wall to be designed and detailed in accordance with NZS 1720.1:2022 Timber Structures Part 1 - Design methods or NZ3604



Glass Cleaning and Maintenance

Architectural glass products must be properly cleaned during the construction period so visual and aesthetic clarity are maintained. Because glass can be permanently damaged if improperly cleaned, glass producers and fabricators recommend strict compliance with the following procedures.

First, determine whether the glass is clear, tinted or reflective. Surface damage is more noticeable on reflective glass compared with the other glass products. If the reflective coated surface is exposed, either on the exterior or interior, special care must be taken when cleaning, as scratches can result in coating removal and a visible change in light transmittance. Cleaning tinted and reflective glass in direct sunlight should be avoided. Cleaning should begin at the top of the building and continue to the lower levels.

Commence cleaning by soaking the glass surfaces with clean water and a soap solution to loosen dirt or debris. Then, using a mild, non-abrasive commercial window washing solution, uniformly apply the solution to the glass surfaces with a non-abrasive applicator and follow with a squeegee to remove all of the cleaning solution from the glass surface.

Ensure that no metal parts of the cleaning equipment touch the glass surface and that no abrasive particles are trapped between the glass and the cleaning materials. All water and cleaning solution residue should be dried from the window gaskets, sealants and frames.

Scratches and Metal Scrapers

Scratches can occur from hard pointed objects or poor handling, but most often occurs from the careless removal of foreign matter from the glass surface.

Mortar splatter and paint are common offenders and efforts to remove after hardening almost always lead to surface damage. It is essential that the foreign materials are removed before they harden. Better still, if construction work continues after glazing, that the glazed areas are protected by adhesive plastic films or suitable tarpaulins or covers.

One of the common mistakes made by non-glass trades people, including glass cleaning contractors, is the use of razor blades or other metal scrapers on a large portion of the glass surface. Using large blades to scrape a window clean carries considerable risk of causing damage to the glass.

The glass industry, fabricators, distributors and installers neither condones nor recommends any scraping of glass surfaces with metal blades or knives. Such scraping usually permanently damages or scratches the glass surfaces. When paint or other construction materials cannot be removed with normal cleaning procedures, a new 25mm razor blade may have to be used. The razor blade should be used on small spots only. Cleaning should be done in one direction only. Never scrape in a back and forth motion as this could trap particles under the blade that could scratch the glass.

Blades or scrapers can dislodge "pickup" on toughened glass. There are fine particles of glass that are fused on to the surface during toughening. Once dislodged they can scratch the glass.

Glass Cleaning, Do's and Don'ts

DO NOT..

- Do Not - Use Scrapers of any type or size on a Glass surface
- Do Not - Leave building dirt or residues to remain on Glass for a period of time.
- Do Not - Begin cleaning glass until you have identified the surface type.
- Do Not - Clean Glass surfaces in direct sunlight.
- Do Not - Allow dirty water or cleaning residues to remain on the Glass.
- Do Not - Begin cleaning before rinsing off a loose residues.
- Do Not - Use abrasive cleaning solutions, materials or solvents.
- Do Not - Allow metal parts of the cleaning equipment to come in contact with the Glass.
- Do Not - Trap abrasive particles between the cleaning material and the Glass.

DO...

- Clean glass promptly when dirt or building residues appear.
- Determine glass surface type.
- Exercise special care when cleaning coated surfaces.
- Avoid cleaning glass surfaces in direct sunlight.
- Start cleaning at the top of a building, then continue to lower levels.
- Soak the glass surface in a clean soapy solution before cleaning.
- Use a mild non abrasive commercial cleaner.
- Use a squeegee to remove all cleaning solution.
- Try your procedures on a small window and check.
- Caution other trades re the care and protection of the glass surfaces.

**Residues of surface grit may be present from the toughening production process.
These grit particles must not be dragged across the surface.
NEVER use Metal Scrapers**

All above reprinted with permission from Metro Glass Tech



Powder Coating Installation Care

Warning re use of solvents:

- In some cases strong solvents are recommended for thinning various types of paints and also for cleaning up mastics and sealants.
- These can be harmful to the extended life of the powder coated surface, and must not be used for cleaning purposes.
- It is important to note that the damage will not be visible immediately and may take up to 12 months to develop.

If paint splashes or sealants and mastics need to be removed then the following may be safely used:
Methylated Spirits, Ethyl Alcohol, Isopropanol or preferably a mild detergent in warm water.

Joinery Protection during Installation:

All the activity on a construction site means that your powder coated items may get knocked or scratched, splattered with mortar, plaster, textured coating or paint during the later stages of construction.

Please ensure that all powder coated articles are masked or covered at this time. It is far easier to prevent accidents than to try and correct them. Should your joinery receive mortar or paint splashes see that these are removed before cure and follow the instructions contained in this brochure.

Typical sticker used to warn other trades of the need to protect and mask off powder coated joinery (applies to anodised joinery also)

"IMPORTANT ALL TRADES"
This valuable aluminium joinery will suffer permanent damage from: plaster, mortar and paint splashes - Protect if splashes occur - Immediately wash down joinery with water or meths - Do not allow splashes to harden! - Do not use solvents! - Do not remove this label until final clean completed.

This photograph display damage that has occurred on site, post installation. The photo of the masked joinery displays clear signs of damage that could have occurred were it not masked. Please ensure that your joinery is protected right through the entire construction process.



Powder Coating Maintenance

External - Maintenance Program:

To extend the life of external powder coated articles and to comply with warranty requirements for powder coated aluminium joinery, a simple, regular maintenance program must be implemented.

The effects of ultra violet light, atmospheric pollution, dirt, grime and airborne salt deposits will all accumulate over time and must be removed or surface staining and weathering will occur, leading to an unsightly appearance.

For external coatings, cleaning should take place every six months. In areas where pollutants are more prevalent, such as beachfront houses and industrial or geothermal areas, then a cleaning program should be carried out on a more frequent basis ie. every one to three months.

Fences or Balustrades in close proximity to swimming pools must be washed down every six months, to clean off chlorine and salt deposits.

Cleaning your powder coating:

1. Carefully remove any loose surface deposits with a wet sponge.
2. Use a soft brush (non abrasive) and a mild household detergent (do not use solvents) in warm water, remove dust, salt and other deposits.
3. Rinse off with clean fresh water.



Restoring weathered or scratched surfaces:

Repair of Scuffed or Scratched surfaces

Dulux Spray Cans are available in all colour card colours.

Repair of Small Scratches or Chips.

Dulux Dabsticks are ideally suited for the repair of small scratches.

Dabsticks may not be available in all colour card colours.

Repair of Weathered areas .

Dulux Gloss Up is a light to medium cutting cream ideally suited for gloss restoration and has been specifically designed for this purpose. Gloss Up contains no waxes or silicone and is a one step system.



Contact Dulux Powder Coatings , ph 0064 9 441 8244

Juralco Edgetec® JH Clamp Balustrade System

Stainless Steel Care and Maintenance

Care and Maintenance of Stainless Steel

Introduction

Stainless steels are selected for applications where their inherent corrosion resistance, strength and aesthetic appeal are required. However, dependent on the service conditions, stainless steels will stain and discolour due to surface deposits and so cannot be assumed to be completely maintenance-free. In order to achieve maximum corrosion resistance and aesthetic appeal, the surface of the stainless steel must be kept clean. Provided the grade of stainless steel and the surface finish are correctly selected, and cleaning schedules carried out on a regular basis, good performance and long service life will result.

For the correct selection of a Stainless Steel grade, with respect to Location, see Table below.

Factors affecting maintenance

Surface contamination and the formation of deposits on the surface of the stainless steel must be prevented. These deposits may be minute particles of iron or rust generated during construction. Industrial and even naturally occurring atmospheric conditions can produce deposits which can be equally corrosive, e.g. salt deposits from marine conditions.

Working environments can also provide aggressive conditions such as heat and humidity in swimming pool buildings. These conditions can result in surface discolouration of stainless steels and so maintenance on a more frequent basis may be required.

Modern processes use many cleaners, sterilizers and bleaches for hygienic purposes. Proprietary solutions, when used in accordance with makers' instructions, should be safe but if used incorrectly (e.g. warm or concentrated), may cause discolouration or corrosion on stainless steels. Strong acid solutions are sometimes used to clean masonry and tiling of buildings. These acids should never be used where contact with metals, including stainless steel, is possible. If this happens, the acid solution must be removed immediately, followed by dilution and rinsing with clean water.

Maintenance programme

With care taken during fabrication and installation, cleaning before 'hand-over' should not present any problems. More attention may be required if the installation period has been prolonged or hand-over delayed. Where surface contamination is suspected, immediate cleaning after site fixing should avoid problems later.

The frequency of cleaning is dependent on the application. This may vary from once to four times a year for external applications, Recommendations on cleaning frequencies in architectural applications are shown below.

Cleaning frequency

Reccommended Cleaning for various grades of Stainless Steel		
Location	304 Grade	316 Grade
Surbarban or Rural	Clean at 6-12mth intervals or as necessary	
Industrial or Urban	Clean at 3-6mth intervals	Clean at 6-12mth intervals
Coastal or Marine	Not recommended	